The Koide Quadratic Cone as a Lorentzian Determinant

Vieta dynamics and a ternary defect-one counting problem

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Abstract

Writing the charged-lepton Koide relation in square-root mass coordinates produces a quadratic cone of signature $(1, 2)$. We give an explicit orthogonal identification of this cone with the determinant-zero locus in $\mathrm{Sym}_2(\mathbb{R})$. In the exact period convention $\Pi = [Z \mid I_2]$, $Z = \left(\begin{smallmatrix} 6\mu & \tau \\ \beta & \mu \end{smallmatrix}\right)$, used in a recent proposed complex-torus construction on S^6, the same map identifies the Koide cone with the zero-discriminant locus in the ambient imaginary-coefficient space. This coefficient-space calculation neither assumes nor establishes the claimed global construction. The arithmetic part is independent of it. For

$$\Delta(x, y, z) = 4(xy + xz + yz) - (x^2 + y^2 + z^2),$$

we study ordered positive solutions of $\Delta = -1$ reduced by the Vieta wall $z \geq 4(x + y)$. An exact integral change of variables identifies them with a chamber on

$$a^2 - 3q^2 - 18n^2 = -2, \qquad z = 4a + 17n.$$

The three Vieta substitutions are integral reflections of the Lorentzian form Δ; on unordered positive solutions they form a forest in which every component has one reduced root. The two boundary faces of the ternary chamber are completely described by negative-Pell recurrences and converge to the two null directions bounding the archimedean sector. The local densities, the archimedean chamber factor, the Gorodnik–Oh–Shah sector hypotheses, and the normalization against Borovoi's Tamagawa measure are all made explicit. A 2-adic sign-reversing, measure-preserving bijection proves that Borovoi's adelic coefficient is one, yielding the unconditional explicit main term. An exact HNF parametrization by ideals of $\mathbb{Z}[\sqrt{3}]$, unique unit-orbit descent, and a sharp floor formula identify every counted point with multiplicity one. Direct verification of Benoist–Oh effective well-roundedness then gives $R(X) = C_0 X + O(X^{1-\eta})$ for some unspecified $\eta > 0$. The same shell is identified exactly with a finite union of primitive hyperbolic conjugacy classes in a cocompact quaternionic lattice, but its standard radial height is $7a + 15\sqrt{2}\, n$, not the sharp height $4a + 17n$. Exact enumeration through height 10^8 gives $R(10^8) = 3{,}536{,}002$. The primary-source audit finds no theorem proving the pointwise square-root discrepancy for the exact five-wall cutoff; that stronger estimate remains open.

Status. This is a mathematical draft for independent checking. It makes no claim that the Koide relation is a law of particle physics, no claim about the existence of the proposed complex structure on S^6, and no Yang–Mills claim. The exact algebraic identities, parametrization, theorem audit, and bounded census are stated with their verification data in the canonical source modules included below.

1 The Koide form and its Lorentzian model

For a triple of nonnegative masses m_1, m_2, m_3, put $r_i = \sqrt{m_i}$. The Koide relation may be written

$$m_1 + m_2 + m_3 = \frac{2}{3}\left(\sqrt{m_1} + \sqrt{m_2} + \sqrt{m_3}\right)^2, \tag{1.1}$$

which was introduced in the charged-lepton context by Koide (1983). The circle interpretation of Kocik (2012) supplies a separate Euclidean realization. Here the relevant object is the homogeneous quadratic form

$$Q_K(r_1, r_2, r_3) := \frac{2}{3}(r_1 + r_2 + r_3)^2 - (r_1^2 + r_2^2 + r_3^2). \tag{1.2}$$

Thus (1.1) is $Q_K(r) = 0$.

Define the orthogonal Fourier coordinates

$$\mathcal{F}(r_1, r_2, r_3) = \left(\frac{r_1 + r_2 + r_3}{\sqrt{3}}, \frac{r_1 - r_3}{\sqrt{2}}, \frac{r_1 + r_3 - 2r_2}{\sqrt{6}}\right) =: (h, x, y), \tag{1.3}$$

and the symmetric-matrix embedding

$$\iota(h, x, y) = \begin{pmatrix} h + y & x \\ x & h - y \end{pmatrix}. \tag{1.4}$$

Theorem 1.1 (Orthogonal determinant model)**.** *The map $\mathcal{F}$ is orthogonal and*

$$Q_K(r) = h^2 - x^2 - y^2 = \det \iota(h, x, y). \tag{1.5}$$

Its inverse is

$$r_1 = \frac{h}{\sqrt{3}} + \frac{x}{\sqrt{2}} + \frac{y}{\sqrt{6}}, \tag{1.6}$$

$$r_2 = \frac{h}{\sqrt{3}} - \frac{2y}{\sqrt{6}}, \tag{1.7}$$

$$r_3 = \frac{h}{\sqrt{3}} - \frac{x}{\sqrt{2}} + \frac{y}{\sqrt{6}}. \tag{1.8}$$

For $r \in \mathbb{R}^3_{\geq 0} \setminus \{0\}$, the Koide equation is equivalent to the assertion that $\iota(\mathcal{F}(r))$ is positive semidefinite of rank one. Equivalently, the Euclidean angle between r and $(1,1,1)/\sqrt{3}$ is $\pi/4$.

Proof. The three rows of the matrix in (1.3) form an orthonormal basis. Hence

$$r_1^2 + r_2^2 + r_3^2 = h^2 + x^2 + y^2, \qquad (r_1 + r_2 + r_3)^2 = 3h^2,$$

and substitution in (1.2) gives (1.5). Transposition of the orthogonal matrix gives (1.6)–(1.8). If $r \in \mathbb{R}^3_{\geq 0} \setminus \{0\}$, then $h > 0$. On the null cone, $h^2 = x^2 + y^2$, so ι has trace $2h > 0$, determinant zero, and eigenvalues $2h, 0$. Finally,

$$\cos \angle\left(r, \frac{(1,1,1)}{\sqrt{3}}\right) = \frac{h}{\sqrt{h^2 + x^2 + y^2}} = \frac{1}{\sqrt{2}}.$$

All implications reverse. □

Corollary 1.2 (Exact phase sector)**.** *Let $r \in \mathbb{R}^3$ satisfy $Q_K(r) = 0$, and suppose that the first Fourier coordinate in (1.3) satisfies $h > 0$. There is a unique $\theta \in \mathbb{R}/(2\pi\mathbb{Z})$ such that*

$$r_1 = \frac{h}{\sqrt{3}}\left(1 + \sqrt{2}\cos\left(\theta - \frac{\pi}{6}\right)\right), \tag{1.9}$$

$$r_2 = \frac{h}{\sqrt{3}}\left(1 + \sqrt{2}\cos\left(\theta + \frac{\pi}{2}\right)\right), \tag{1.10}$$

$$r_3 = \frac{h}{\sqrt{3}}\left(1 + \sqrt{2}\cos\left(\theta - \frac{5\pi}{6}\right)\right). \tag{1.11}$$

Moreover, $r_1, r_2, r_3 \geq 0$ if and only if

$$\theta \in \left[\frac{\pi}{12}, \frac{\pi}{4}\right] \cup \left[\frac{3\pi}{4}, \frac{11\pi}{12}\right] \cup \left[\frac{17\pi}{12}, \frac{19\pi}{12}\right] \pmod{2\pi}. \tag{1.12}$$

At each endpoint exactly one coordinate vanishes, and the ratio of the two nonzero coordinates is $2 + \sqrt{3}$ or $(2 + \sqrt{3})^{-1} = 2 - \sqrt{3}$.

Proof. The null equation and $h > 0$ give $x = h\cos\theta$, $y = h\sin\theta$ for a unique $\theta \in \mathbb{R}/(2\pi\mathbb{Z})$. Substitution in (1.6)–(1.8), followed by the three elementary angle-addition identities, gives (1.9)–(1.11).

The inequality $1 + \sqrt{2}\cos\phi \geq 0$ fails precisely for $3\pi/4 < \phi < 5\pi/4$ modulo 2π. Applied to the three phases in (1.9)–(1.11), the three excluded open arcs are

$$\left(\frac{11\pi}{12}, \frac{17\pi}{12}\right), \qquad \left(\frac{\pi}{4}, \frac{3\pi}{4}\right), \qquad \left(\frac{19\pi}{12}, 2\pi\right) \cup \left(0, \frac{\pi}{12}\right).$$

Their complement is exactly (1.12). If, for example, $r_3 = 0$, the unchanged equation $Q_K(r) = 0$ becomes

$$r_1^2 - 4r_1r_2 + r_2^2 = 0,$$

so $r_1/r_2 = 2 \pm \sqrt{3}$. Cyclic symmetry proves the assertion at every endpoint. □

1.1 The canonical period discriminant as an ambient model

The recent online manuscript Alpöge (2026) uses period functions τ, μ, β and the exact convention

$$Z = \begin{pmatrix} 6\mu & \tau \\ \beta & \mu \end{pmatrix}, \qquad \Pi = [Z \mid I_2]. \tag{1.13}$$

Write

$$t = \operatorname{Im}\tau, \qquad m = \operatorname{Im}\mu, \qquad \varrho = \operatorname{Im}\beta.$$

Relative to the period-lattice basis $(\widehat{\gamma}, \widehat{u}, \widehat{w}, \widehat{\delta})$ and the interleaved real codomain order $(\operatorname{Re} z_1, \operatorname{Im} z_1, \operatorname{Re} z_2, \operatorname{Im} z_2)$, the realification of (1.13) is

$$P(\tau, \mu, \beta) = \begin{pmatrix} 6\operatorname{Re}\mu & \operatorname{Re}\tau & 1 & 0 \\ 6m & t & 0 & 0 \\ \operatorname{Re}\beta & \operatorname{Re}\mu & 0 & 1 \\ \varrho & m & 0 & 0 \end{pmatrix}. \tag{1.14}$$

For comparison with the Lorentz cone, define the auxiliary symmetric coefficient matrix

$$S(t, m, \varrho) := \begin{pmatrix} t & \sqrt{6}\,m \\ \sqrt{6}\,m & \varrho \end{pmatrix}. \tag{1.15}$$

The matrix S represents the same quadratic discriminant, but it is not the canonical 4×4 real period matrix P.

Proposition 1.3 (Exact canonical coefficient dictionary)**.** *The real-linear map*

$$\mathcal{K}(h,x,y)=\left(h+y,\frac{x}{\sqrt{6}},h-y\right)=(t,m,\varrho) \tag{1.16}$$

is an isomorphism from $\mathbb{R}^3$ *to the ambient imaginary-coefficient space, with inverse*

$$\mathcal{K}^{-1}(t,m,\varrho)=\left(\frac{t+\varrho}{2},\sqrt{6}\,m,\frac{t-\varrho}{2}\right). \tag{1.17}$$

For every choice of the real parts of τ,μ,β*,*

$$\det P(\tau,\mu,\beta)=\det S(t,m,\varrho)=t\varrho-6m^2=-\det \operatorname{Im} Z. \tag{1.18}$$

Consequently, if $(t,m,\varrho)=\mathcal{K}(\mathcal{F}(r))$*, then*

$$\det P(\tau,\mu,\beta)=\det S\big(\mathcal{K}(\mathcal{F}(r))\big)=Q_K(r). \tag{1.19}$$

For $r\in\mathbb{R}^3_{\geq 0}\setminus\{0\}$ *on the Koide cone, the symmetric matrix* $S\big(\mathcal{K}(\mathcal{F}(r))\big)=\iota(\mathcal{F}(r))$ *is positive semidefinite of rank one.*

Proof. Adding and subtracting the first and third entries of (1.16), and multiplying the second by $\sqrt{6}$, gives (1.17). Expanding (1.14) along its last two columns gives

$$\det P=-\det\begin{pmatrix}6m & t\\ \varrho & m\end{pmatrix}=t\varrho-6m^2.$$

The same expression is the determinant of (1.15), and substitution from (1.16) gives

$$(h+y)(h-y)-6\left(\frac{x}{\sqrt{6}}\right)^2=h^2-x^2-y^2=Q_K(r).$$

The final assertion is Theorem 1.1 with $S\big(\mathcal{K}(h,x,y)\big)=\iota(h,x,y)$. □

Corollary 1.4 (Rank-one square parametrization)**.** *The map* $\mathcal{K}$ *identifies the future Lorentz null cone* $\{h^2=x^2+y^2,\ h>0\}$ *with the nonzero positive-semidefinite rank-one locus in* $\mathrm{Sym}_2(\mathbb{R})$*. Equivalently, every point of that coefficient locus has the form*

$$(t,m,\varrho)=\left(u^2,\frac{uv}{\sqrt{6}},v^2\right),\qquad (u,v)\in\mathbb{R}^2\setminus\{(0,0)\}, \tag{1.20}$$

and (u,v) *is unique up to simultaneous sign. Under the inverse coefficient map,*

$$(h,x,y)=\left(\frac{u^2+v^2}{2},uv,\frac{u^2-v^2}{2}\right). \tag{1.21}$$

The sublocus arising from nonnegative r_i *is exactly the phase sector in Corollary 1.2; positive semidefiniteness alone does not impose the three mass-coordinate inequalities.*

Proof. A nonzero real symmetric matrix is positive semidefinite of rank one if and only if it is ww^{T} for a nonzero column $w=(u,v)^{\mathsf{T}}$; the factor w is unique up to $w\mapsto -w$. Comparing ww^{T} with (1.15) gives (1.20), and (1.17) gives (1.21). Conversely the latter triple has $h>0$ and $h^2-x^2-y^2=0$. The final statement follows from Corollary 1.2. □

Corollary 1.5 (The Koide cone is not a family of regular tori)**.** *The identification in Proposition 1.3 is only a real-linear identification of ambient coefficient spaces and their quadratic discriminants. It supplies neither the real parts of the periods nor the holomorphic and equivariance conditions of the proposed period family. At every point of the Koide cone, every matrix of the canonical form (1.14) with coefficient triple (1.16) has determinant zero. Its columns therefore do not span $\mathbb{R}^4$, do not generate a full lattice, and do not define a regular compact complex two-torus.*

Remark 1.6 (Separation from the positive radial cusp)*.* The audited period calculation associated with Alpöge (2026) records holomorphic cusp germs μ, h_{per}, b at $t_c = 0$, with $s = iT$, $\tau = h_{\mathrm{per}} + s$, and $b = \beta + \tau$ on the positive radial specialization $t_c = e^{-2\pi T}$. Write

$$\mu_T = \alpha_T + i m_T, \qquad h_{\mathrm{per},T} = \eta_T + i H_T, \qquad b_T = \xi_T + i B_T,$$

and set

$$c_T = \xi_T - \eta_T, \qquad L_T = T + H_T, \qquad q_T = L_T - B_T, \qquad D_T = L_T q_T + 6m_T^2.$$

Then the imaginary-coefficient triple is exactly

$$(\operatorname{Im}\tau_T, \operatorname{Im}\mu_T, \operatorname{Im}\beta_T) = (L_T, m_T, -q_T). \tag{1.22}$$

Its exact realification is

$$P_T = \begin{pmatrix} 6\alpha_T & \eta_T & 1 & 0 \\ 6m_T & L_T & 0 & 0 \\ c_T & \alpha_T & 0 & 1 \\ -q_T & m_T & 0 & 0 \end{pmatrix}, \qquad \det P_T = -D_T. \tag{1.23}$$

Indeed, (1.18) and (1.22) give

$$\det P_T = L_T(-q_T) - 6m_T^2 = -D_T.$$

If $m = \operatorname{Im}\mu(0)$, $H = \operatorname{Im}h_{\mathrm{per}}(0)$, and $B = \operatorname{Im}b(0)$, then

$$m_T = m + O(e^{-2\pi T}), \quad H_T = H + O(e^{-2\pi T}), \quad B_T = B + O(e^{-2\pi T}).$$

Direct multiplication, without suppressing the constant term, gives

$$D_T = T^2 + (2H - B)T + H(H - B) + 6m^2 + O(Te^{-2\pi T}), \qquad \det P_T \sim -T^2. \tag{1.24}$$

In particular $D_T > 0$ for all sufficiently large T, so these radial matrices are invertible and lie strictly off the Koide discriminant hypersurface $\{\det P = 0\}$; their period-lattice covolume is then D_T. More precisely, their auxiliary symmetric matrix is

$$S_T = \begin{pmatrix} L_T & \sqrt{6}\, m_T \\ \sqrt{6}\, m_T & -q_T \end{pmatrix}, \qquad \det S_T = -D_T < 0,$$

and therefore has signature $(1, 1)$ for all sufficiently large T. This is different not merely from the zero determinant on the Koide locus but also from its positive-semidefinite rank-one sign chamber. This exact separation uses the recorded cusp formulas only; it is not an independent proof of the proposed global S^6 construction.

2 Exact arithmetic of the Koide defect-one shell

This section isolates an arithmetic result that does not use the existence of the proposed complex structure on S^6, any period-map assertion, or any Yang–Mills interpretation. All coordinates, chambers, local factors, and normalizations used in the final constant are fixed explicitly below.

For $r = (r_1, r_2, r_3)$, define the Koide quadratic form

$$Q_K(r_1, r_2, r_3) := \frac{2}{3}(r_1 + r_2 + r_3)^2 - (r_1^2 + r_2^2 + r_3^2). \tag{2.1}$$

For integral coordinates put

$$\Delta(x, y, z) := 4(xy + xz + yz) - (x^2 + y^2 + z^2). \tag{2.2}$$

Expanding (2.1) gives the exact identity

$$\Delta(x, y, z) = 3Q_K(x, y, z). \tag{2.3}$$

Thus the equation studied here, $\Delta = -1$, is the nonzero level $Q_K = -1/3$; it is not an affine translate of the null cone $Q_K = 0$.

2.1 Vieta reduction and exact ternary coordinates

Regarded as a monic quadratic equation in z, the defect equation is

$$z^2 - 4(x + y)z + x^2 + y^2 - 4xy - 1 = 0. \tag{2.4}$$

Consequently the other root is

$$z' = 4(x + y) - z. \tag{2.5}$$

The analogous substitutions in x and y are integral involutions preserving Δ.

Let e_1, e_2, e_3 be the standard basis and let B_K be the integral bilinear form with Gram matrix

$$A_K = \begin{pmatrix} -1 & 2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & -1 \end{pmatrix}, \qquad B_K(u, v) = u^{\mathsf{T}} A_K v. \tag{2.6}$$

Thus $\Delta(u) = B_K(u, u)$. The eigenvalue of A_K on $\mathbb{R}(1, 1, 1)$ is 3, and its eigenvalue on the orthogonal complement is -3, so this integral presentation has signature $(1, 2)$.

Proposition 2.1 (Vieta moves are Lorentzian reflections)**.** *For $i = 1, 2, 3$, define*

$$s_i(u) = u - \frac{2B_K(u, e_i)}{B_K(e_i, e_i)} e_i. \tag{2.7}$$

Then $s_i \in \mathrm{O}(B_K, \mathbb{Z})$, $s_i^2 = 1$, and $\det s_i = -1$. Explicitly,

$$s_1(x, y, z) = (-x + 4y + 4z, y, z), \tag{2.8}$$
$$s_2(x, y, z) = (x, -y + 4x + 4z, z), \tag{2.9}$$
$$s_3(x, y, z) = (x, y, -z + 4x + 4y). \tag{2.10}$$

Consequently these are exactly the three Vieta root substitutions for $\Delta = -1$, now identified as reflections of the unchanged Lorentzian lattice $(\mathbb{Z}^3, B_K)$.

Proof. Since $B_K(e_i, e_i) = -1$, formula (2.7) is integral and direct evaluation gives (2.8)– (2.10). The standard reflection calculation gives

$$B_K(s_i(u), s_i(v)) = B_K(u, v), \qquad s_i^2 = 1.$$

The map fixes the hyperplane $e_i^\perp$ pointwise and sends e_i to $-e_i$, hence has determinant -1. Finally, when the equation is viewed as a monic quadratic in one coordinate, the displayed replacement is the sum of its two roots minus the original root, proving the last assertion. □

Proposition 2.2 (Strict Vieta reduction)**.** *Let $0 < x \le y \le z$ be integers satisfying $\Delta(x, y, z) = -1$. If $z < 4(x + y)$, then the other root in (2.5) satisfies $0 < z' < y$. If $z \ge 4(x + y)$, then $z' \le 0$.*

Proof. In the first case $z' > 0$ follows immediately from (2.5). Let $P(Z)$ be the polynomial on the left of (2.4). Since $x \le y$,

$$P(y) = x^2 - 8xy - 2y^2 - 1 \le xy - 8xy - 2y^2 - 1 < 0.$$

But $P(y) = (y - z)(y - z')$, and $y - z \le 0$; hence $y - z' > 0$, so $z' < y$. The second assertion again follows directly from (2.5). □

Define the reduced-root set and its height count by

$$\mathcal{R} := \{(x, y, z) \in \mathbb{Z}^3_{>0} : x \le y \le z,\ \Delta(x, y, z) = -1,\ z \ge 4(x + y)\}, \tag{2.11}$$

$$R(X) := \#\{(x, y, z) \in \mathcal{R} : z \le X\}. \tag{2.12}$$

Proposition 2.2 shows that the wall $z \ge 4(x + y)$ is exactly the terminal side of the decreasing Vieta move after the coordinates have been ordered.

Proposition 2.3 (The positive Vieta graph is a rooted forest)**.** *Let $\mathcal{S}^+$ be the set of unordered triples of positive integers satisfying $\Delta = -1$. Join two vertices in a simple graph when one is obtained from the other by a Vieta reflection s_i and the reflected coordinate is positive. Every connected component of this graph is a tree with exactly one root, and the roots are precisely the unordered triples whose ordered representatives lie in $\mathcal{R}$. Equivalently, every positive integral solution has a unique reduced representative, and it is obtained by repeatedly ordering the coordinates and reflecting the largest coordinate whenever it lies below the terminal wall.*

Proof. First, no Vieta reflection fixes a point of $\Delta = -1$. Indeed, a fixed reflection in a coordinate w, with the other two coordinates denoted by u, v, would give $w = 2(u + v)$, and substitution in the unchanged form gives

$$\Delta(u, v, 2(u + v)) = 3(u^2 + 4uv + v^2) > 0.$$

Thus the sum of the three coordinates changes strictly across every edge.

Now order a vertex as $0 < x \le y \le z$. In fact $y < z$, because $y = z$ would give

$$\Delta(x, y, y) = -x^2 + 8xy + 2y^2 > 0.$$

Reflecting either of the two smaller coordinates produces a new coordinate strictly larger than z:

$$4(y + z) - x > z, \qquad 4(x + z) - y > z.$$

If $z < 4(x+y)$, Proposition 2.2 says that reflection in the largest coordinate produces $0 < z' < y$ and strictly decreases the coordinate sum. If $z \ge 4(x + y)$, that reflection is nonpositive and

hence is not an edge of $\mathcal{S}^+$. Therefore every nonterminal vertex has exactly one neighbour of smaller coordinate sum, whereas a terminal vertex has none.

Iteration of the unique decreasing move terminates because the coordinate sum is a positive integer. The endpoint is represented by an element of $\mathcal{R}$. The graph has no cycle: on a putative simple cycle choose a vertex of largest coordinate sum; its two neighbours on the cycle would be two distinct smaller neighbours, contrary to uniqueness. Thus each component is a tree. A tree in which following the unique decreasing edge always terminates has exactly one terminal vertex, proving both uniqueness and the characterization of the roots. □

For an ordered triple define

$$n = z - 4(x + y), \qquad q = y - x, \qquad a = x + y - 4n, \tag{2.13}$$

and let

$$\mathcal{T} := \left\{ (a, q, n) \in \mathbb{Z}^3 : \begin{array}{c} a > 0,\ q \geq 0,\ n \geq 0,\ a + 4n > q, \\ a^2 - 3q^2 - 18n^2 = -2 \end{array} \right\}. \tag{2.14}$$

Theorem 2.4 (Integral ternary-coordinate bijection)**.** *The map in* (2.13) *restricts to a bijection* $\Phi : \mathcal{R} \to \mathcal{T}$*. Its inverse is*

$$(a, q, n) \longmapsto \left(\frac{a + 4n - q}{2}, \frac{a + 4n + q}{2}, 4a + 17n \right). \tag{2.15}$$

For arbitrary integral (x, y, z) *and the associated* (a, q, n)*,*

$$2\Delta(x, y, z) = a^2 - 3q^2 - 18n^2, \qquad z = 4a + 17n. \tag{2.16}$$

In particular,

$$R(X) = \#\{(a, q, n) \in \mathcal{T} : 4a + 17n \leq X\}. \tag{2.17}$$

Proof. Put $p = x + y$. Since $q = y - x$, one has $2x = p - q$ and $2y = p + q$, and direct substitution into (2.2) gives

$$2\Delta = p^2 - 3q^2 + 8pz - 2z^2.$$

The definition $n = z - 4p$ gives $z = 4p + n$, whence

$$2\Delta = p^2 - 3q^2 - 8pn - 2n^2.$$

Finally $a = p - 4n$, equivalently $p = a + 4n$, gives both identities in (2.16).

Suppose first that $(x, y, z) \in \mathcal{R}$. Then $q, n \geq 0$, while $a + 4n = p > q$ because $p - q = 2x > 0$. The defect equation and (2.16) give $a^2 - 3q^2 - 18n^2 = -2$. If $n = 0$, then $a = p > 0$. If $n \geq 1$, then

$$a^2 = 3q^2 + 18n^2 - 2 \geq 18n^2 - 2 \geq 16n^2.$$

Were $a \leq 0$, this inequality would force $a \leq -4n$, contrary to $a + 4n > q \geq 0$. Hence $a > 0$, and the forward map lands in $\mathcal{T}$.

Conversely, reducing the ternary equation modulo 2 gives $a \equiv q \pmod 2$, so all three coordinates in (2.15) are integral. Put $p = a + 4n$. The chamber inequality gives

$$x = \frac{p - q}{2} > 0, \qquad y = \frac{p + q}{2} \geq x.$$

Moreover

$$z = 4a + 17n = 4p + n \geq 4p > p > y.$$

The identities (2.16) now give $\Delta = -1$ and $z - 4(x + y) = n \geq 0$. Thus the inverse map lands in $\mathcal{R}$. The displayed formulae show directly that the two compositions are the identity. The last assertion follows from the height identity in (2.16). □

Proposition 2.5 (Complete Pell classification of the chamber faces)**.** *The integral points of $\mathcal{T}$ on its two coordinate faces are the following two disjoint families.*

On the terminal Vieta wall $n = 0$, define integers a_k, q_k by

$$a_k + q_k\sqrt{3} = (1+\sqrt{3})(2+\sqrt{3})^k \qquad (k \geq 0). \tag{2.18}$$

Then the points of $\mathcal{T}$ with $n = 0$ are exactly $(a_k, q_k, 0)$ for $k \geq 1$, and their reduced roots are

$$\left(\frac{a_k - q_k}{2}, \frac{a_k + q_k}{2}, 4a_k\right). \tag{2.19}$$

The first is $(1, 4, 20)$.

On the symmetry face $q = 0$, define integers A_k, N_k by

$$A_k + N_k\sqrt{18} = (4+\sqrt{18})(17+4\sqrt{18})^k \qquad (k \geq 0). \tag{2.20}$$

Then the points of $\mathcal{T}$ with $q = 0$ are exactly $(A_k, 0, N_k)$ for $k \geq 0$, and their reduced roots are

$$\left(\frac{A_k + 4N_k}{2}, \frac{A_k + 4N_k}{2}, 4A_k + 17N_k\right). \tag{2.21}$$

The first is $(4, 4, 33)$.

In the projectivized real quadric these families tend, respectively, to the two chamber-boundary rays

$$(a, q, n) = \mathbb{R}_{>0}(\sqrt{3}, 1, 0), \qquad (a, q, n) = \mathbb{R}_{>0}(3\sqrt{2}, 0, 1). \tag{2.22}$$

Each family contributes only $O(\log X)$ points to $R(X)$.

Proof. On $n = 0$, the equation of $\mathcal{T}$ is $a^2 - 3q^2 = -2$. Here $q = 0$ is impossible. If $q = 1$, positivity gives $a = 1$, which fails the strict chamber inequality $a > q$. For any positive solution with $q \geq 2$, multiplication by the norm-one unit $2 - \sqrt{3}$ gives

$$a' + q'\sqrt{3} = (a + q\sqrt{3})(2 - \sqrt{3}), \qquad a' = 2a - 3q, \quad q' = 2q - a. \tag{2.23}$$

The unchanged equation implies

$$a < 2q, \qquad a > \frac{3}{2}q, \qquad a > q \quad (q \geq 2),$$

because its squares differ from $4q^2$, $9q^2/4$, and q^2 by, respectively, $q^2 + 2$, $3q^2/4 - 2 > 0$, and $2q^2 - 2 > 0$. Thus $a', q' > 0$ and $q' < q$. Repeated descent reaches $(1, 1)$, and reversing the descent proves (2.18) is the complete positive solution set. The recurrence

$$a_{k+1} = 2a_k + 3q_k, \qquad q_{k+1} = a_k + 2q_k$$

shows inductively that a_k, q_k are both odd. The chamber excludes only $k = 0$, and Theorem 2.4 now gives (2.19).

On $q = 0$, one has $a^2 - 18n^2 = -2$. Necessarily $n > 0$. Direct substitution for $1 \leq n \leq 5$ gives the sole solution $(a, n) = (4, 1)$. If $n \geq 6$, multiplication by the norm-one unit $17 - 4\sqrt{18}$ gives

$$a' + n'\sqrt{18} = (a + n\sqrt{18})(17 - 4\sqrt{18}), \qquad a' = 17a - 72n, \quad n' = 17n - 4a. \tag{2.24}$$

Both new coordinates are positive. Indeed,

$$\frac{a}{n} = \sqrt{18 - \frac{2}{n^2}} > \frac{72}{17} \quad (n \geq 6), \qquad \frac{a}{n} < \sqrt{18} < \frac{17}{4}.$$

Moreover $a > 4n$ and hence $0 < n' < n$. Descent therefore reaches $(4, 1)$, and reversal proves (2.20) is the complete positive solution set. Its recurrence,

$$A_{k+1} = 17A_k + 72N_k, \qquad N_{k+1} = 4A_k + 17N_k,$$

keeps A_k even and N_k odd, so (2.21) follows from the inverse coordinate formula.

Conjugating the two identities defining the sequences shows

$$\frac{a_k}{q_k} \longrightarrow \sqrt{3}, \qquad \frac{A_k}{N_k} \longrightarrow \sqrt{18} = 3\sqrt{2},$$

which proves (2.22). Finally, the dominant factors $2 + \sqrt{3} > 1$ and $17 + 4\sqrt{18} > 1$ give exponential growth in k, whereas the heights in (2.19) and (2.21) are positive linear forms in the displayed coordinates. Hence either face contains only $O(\log X)$ points of height at most X. □

2.2 The affine quadric and its local densities

Put

$$F(a, q, n) := a^2 - 3q^2 - 18n^2, \qquad \mathscr{X} := \{F = -2\} \subset \mathbb{A}^3. \tag{2.25}$$

The matrix of F is $A = \operatorname{diag}(1, -3, -18)$, so $\det A = 54$, and $-(-2)\det A = 108$ is not a square. The integral point $x_0 = (1, 1, 0)$ lies on $\mathscr{X}$.

For every prime p, define

$$N_{p,k} := \#\{(a, q, n) \bmod p^k : F(a, q, n) \equiv -2 \pmod{p^k}\}, \qquad \sigma_p := \lim_{k\to\infty} p^{-2k} N_{p,k}. \tag{2.26}$$

Theorem 2.6 (Exact local densities)**.** *The limits in* (2.26) *exist and are*

$$\sigma_2 = \frac{3}{2}, \qquad \sigma_3 = 2, \qquad \sigma_p = \frac{1 - p^{-2}}{1 - \left(\frac{3}{p}\right) p^{-1}} \quad (p \geq 5). \tag{2.27}$$

Writing χ_{12} for the primitive real character modulo 12, the singular series, with the primes taken in increasing order (equivalently, with Borovoi's convergence-factor prescription), is therefore

$$\mathfrak{S}(\mathscr{X}) := \prod_p \sigma_p = \frac{9}{2} \frac{L(1, \chi_{12})}{\zeta(2)} = \frac{9}{2} \frac{\log(2 + \sqrt{3})}{\sqrt{3}\,\zeta(2)}. \tag{2.28}$$

Proof. We compute the two bad primes first.

For $p = 3$, once $q, n \bmod 3^k$ have been chosen, the equation for a is

$$a^2 \equiv -2 + 3q^2 + 18n^2 \equiv 1 \pmod 3.$$

The two roots modulo 3 are nonsingular, and each has a unique Hensel lift modulo every 3^k. Thus

$$N_{3,k} = 2 \cdot 3^{2k}, \tag{2.29}$$

which gives $\sigma_3 = 2$.

For $p = 2$, reduction modulo 4 leaves exactly two parity branches:

$$(a, q, n) \equiv (0, 0, 1) \pmod 2, \qquad (a, q, n) \equiv (1, 1, 0) \pmod 2. \tag{2.30}$$

Fix $k \geq 4$ and put $j = k-1$. On the first branch write $a = 2A$, $q = 2Q$, and put $r = 3n$. Then

$$F(a,q,n) + 2 = -2(r^2 - 2A^2 + 6Q^2 - 1). \tag{2.31}$$

Hence we must solve

$$r^2 \equiv 1 + 2A^2 - 6Q^2 \pmod{2^j}.$$

An odd unit is a square modulo 2^j, $j \geq 3$, precisely when it is $1 \pmod 8$, in which case it has four roots. Here that condition is equivalent to A, Q both being even. There are 2^{2j-2} such pairs $(A, Q) \bmod 2^j$, four choices of $r \bmod 2^j$, and, because 3 is a unit, two odd choices of $n \bmod 2^k$ above each such r. This branch therefore contributes

$$2^{2j-2} \cdot 4 \cdot 2 = 2^{2k-1} \tag{2.32}$$

solutions modulo 2^k.

On the second branch write $n = 2m$ and set

$$v = \frac{a-q}{2}, \qquad r = \frac{-a+3q}{2}. \tag{2.33}$$

Then r, v have opposite parity, the inverse formula is $a = r + 3v$, $q = r + v$, and

$$F(a,q,n) + 2 = -2(r^2 - 3v^2 + 36m^2 - 1). \tag{2.34}$$

First suppose $v = 2V$ and r is odd. For fixed $V \bmod 2^{j-1}$ and $m \bmod 2^j$, the equation for r is

$$r^2 \equiv 1 + 12V^2 - 36m^2 \pmod{2^j}.$$

Its right side is $1 \pmod 8$ exactly when V, m have the same parity. There are $2 \cdot 2^{j-2} 2^{j-1} = 2^{2j-2}$ such pairs, and four roots r, so this subbranch gives 2^{2j} triples $(r, v, m) \bmod 2^j$.

Now suppose $r = 2R$ and v is odd. The congruence in (2.34), solved for v^2, has an odd square on the left; its right side is $1 \pmod 8$ exactly when R, m have opposite parity. Again there are 2^{2j-2} admissible pairs and four square roots, hence 2^{2j} transformed triples.

It remains to account for the divided-coordinate map. Each class $(r, v) \bmod 2^j$ of opposite parity has exactly two inverse classes $(a, q) \bmod 2^k$: shifting either r or v by 2^j changes both $a = r + 3v$ and $q = r + v$ by 2^j, while shifting both changes neither modulo 2^k. The class $m \bmod 2^j$ gives one even class $n = 2m \bmod 2^k$. Thus the second branch contributes

$$2(2^{2j} + 2^{2j}) = 2^{2k}. \tag{2.35}$$

Combining (2.32) and (2.35) gives

$$N_{2,k} = 3 \cdot 2^{2k-1} \qquad (k \geq 4), \tag{2.36}$$

and hence $\sigma_2 = 3/2$.

For $p \geq 5$, one has $p \nmid 108$. Borovoi's good-prime formula (Borovoi, 2001, Section 1.3) gives

$$\sigma_p = \frac{1 - p^{-2}}{1 - \chi(p) p^{-1}}, \qquad \chi(p) = \left(\frac{-(-2)\det A}{p}\right) = \left(\frac{108}{p}\right) = \left(\frac{3}{p}\right),$$

which proves the remaining formula in (2.27). Since $\chi(p) = \chi_{12}(p)$ for $p \geq 5$,

$$\prod_{p \geq 5} \sigma_p = L(1, \chi_{12}) \prod_{p \geq 5} (1 - p^{-2}) = \frac{3}{2} \frac{L(1, \chi_{12})}{\zeta(2)}.$$

Multiplication by $\sigma_2\sigma_3 = 3$ gives the first equality in (2.28). Finally, the residue classes on which χ_{12} is nonzero give

$$\begin{aligned} L(1,\chi_{12}) &= \sum_{m\geq 0}\left(\frac{1}{12m+1}-\frac{1}{12m+5}-\frac{1}{12m+7}+\frac{1}{12m+11}\right) \\ &= \int_0^1 \frac{1-x^2}{x^4-x^2+1}\,\mathrm{d}x = \frac{\log(2+\sqrt{3})}{\sqrt{3}}, \end{aligned}$$

where the middle equality follows by summing the geometric series and the last by an elementary partial-fraction integration. This completes the local computation. □

2.3 The archimedean chamber factor

The Poincaré-residue gauge form on the whole smooth affine quadric is the unique regular 2-form ω_∞ whose local representatives satisfy

$$\omega_\infty \wedge \mathrm{d}F = \mathrm{d}a \wedge \mathrm{d}q \wedge \mathrm{d}n, \tag{2.37}$$

Explicitly, on the three standard charts it is

$$\omega_\infty = \frac{\mathrm{d}q\wedge\mathrm{d}n}{2a} \quad (a\neq 0), \qquad \omega_\infty = \frac{\mathrm{d}a\wedge\mathrm{d}n}{6q} \quad (q\neq 0), \qquad \omega_\infty = -\frac{\mathrm{d}a\wedge\mathrm{d}q}{36n} \quad (n\neq 0). \tag{2.38}$$

These expressions agree on overlaps after restriction to the tangent bundle of $F=-2$. Since the gradient $(2a,-6q,-36n)$ cannot vanish on that level, the three charts cover $\mathscr{X}$, so ω_∞ is globally defined and nowhere vanishing. Let $m_\infty := |\omega_\infty|$ be its associated positive measure. On the sheet $a>0$, used for the chamber count, the first expression in (2.38) applies. Let

$$\mathcal{C} := \{(a,q,n)\in\mathbb{R}^3 : a>0,\ q\geq 0,\ n\geq 0,\ a+4n>q\}. \tag{2.39}$$

Theorem 2.7 (Exact archimedean factor)**.** *As $X\to\infty$,*

$$m_\infty\{u\in\mathscr{X}(\mathbb{R})\cap\mathcal{C} : 4a+17n\leq X\}\sim I_\infty X, \qquad I_\infty = \frac{1}{2\sqrt{3}}\log\!\left(\frac{3}{2\sqrt{2}}\right). \tag{2.40}$$

Proof. First, the wall $a+4n>q$ removes only a bounded part of the positive orthant. Indeed, if $q\geq a+4n$, then $a\leq q-4n$; using $a^2=3q^2+18n^2-2$ gives

$$3q^2+18n^2-2\leq (q-4n)^2, \qquad \text{hence} \qquad q^2+4qn+n^2\leq 1.$$

Thus the removed real region has bounded m_∞-measure. Its only integral point on $\mathscr{X}$ with $a>0, q,n\geq 0$ is $(a,q,n)=(1,1,0)$.

Introduce polar coordinates in the negative plane by

$$\sqrt{3}\,q = r\cos\theta, \qquad 3\sqrt{2}\,n = r\sin\theta, \qquad 0\leq\theta\leq\frac{\pi}{2}. \tag{2.41}$$

On $F=-2$, one has $a=\sqrt{r^2-2}$ and $r\,\mathrm{d}r = a\,\mathrm{d}a$. The Jacobian in (2.41) therefore gives

$$\mathrm{d}m_\infty = \frac{1}{6\sqrt{6}}\,\mathrm{d}a\,\mathrm{d}\theta. \tag{2.42}$$

At fixed θ, the height boundary is

$$4a+\frac{17}{3\sqrt{2}}\sqrt{a^2+2}\sin\theta = X. \tag{2.43}$$

Let $A_X(\theta)$ be its nonnegative solution when one exists, and put it equal to zero otherwise. Then $0 \le A_X(\theta)/X \le 1/4$, and for every fixed θ,

$$\frac{A_X(\theta)}{X} \longrightarrow \frac{1}{4 + \frac{17}{3\sqrt{2}} \sin\theta}.$$

Dominated convergence and (2.42) give

$$I_\infty = \frac{1}{6\sqrt{6}} \int_0^{\pi/2} \frac{\mathrm{d}\theta}{4 + \frac{17}{3\sqrt{2}} \sin\theta}. \tag{2.44}$$

For completeness, set $t = \tan(\theta/2)$. If $\alpha = 2\sqrt{2}/3$ and $\beta = 3/(2\sqrt{2})$, then $\alpha\beta = 1$, and

$$\int_0^{\pi/2} \frac{\mathrm{d}\theta}{4 + \frac{17}{3\sqrt{2}} \sin\theta} = \frac{1}{2} \int_0^1 \frac{\mathrm{d}t}{(t+\alpha)(t+\beta)} = 3\sqrt{2} \log\!\left(\frac{3}{2\sqrt{2}}\right).$$

Substitution in (2.44) proves (2.40). □

2.4 A genuine norm and the homogeneous counting theorem

Lemma 2.8 (A genuine norm realizing the height)**.** *Define*

$$\|(a,q,n)\|_* := \max\{|4a + 17n|, |q|, |a|\}. \tag{2.45}$$

Then $\|\cdot\|_*$ is a norm on $\mathbb{R}^3$, and on the chamber $\mathcal{C}$ it satisfies

$$\|(a,q,n)\|_* = 4a + 17n. \tag{2.46}$$

For every integer $X \ge 0$,

$$R(X) = \#\{u \in \mathscr{X}(\mathbb{Z}) \cap \mathcal{C} : \|u\|_* < X + 1\}. \tag{2.47}$$

Proof. The linear map

$$(a,q,n) \longmapsto (4a + 17n, q, a)$$

has determinant -17. Thus (2.45) is the pullback of the supremum norm by an invertible linear map and is a genuine norm. If $(a,q,n) \in \mathcal{C}$, then

$$0 < a < 4a + 17n, \qquad 0 \le q < a + 4n < 4a + 17n,$$

which proves (2.46). Since $4a+17n$ is integral on $\mathscr{X}(\mathbb{Z})$, equation (2.47) follows from (2.17). □

Lemma 2.9 (Verification of the Gorodnik–Oh–Shah hypotheses)**.** *Let $\mathbf{G} = \mathrm{Spin}(F)$, acting through its standard three-dimensional orthogonal representation on $W = \mathbb{R}^3$, and let $x_0 = (1,1,0)$. Then*

$$\mathscr{X}(\mathbb{R}) = \mathbf{G}(\mathbb{R})^\circ x_0.$$

This orbit is a closed affine symmetric variety; $\mathbf{G}$ is connected and absolutely almost simple over $\mathbb{Q}$, is isotropic over $\mathbb{R}$, and acts irreducibly on W over $\mathbb{R}$; and the stabilizer $\mathbf{H} = \mathrm{Stab}_{\mathbf{G}}(x_0)$ has no nontrivial $\mathbb{Q}$-characters. The cone $\mathcal{C}$ in (2.39) is admissible and generic for (Gorodnik et al., 2009, Theorem 1.6).

Proof. The form F has signature $(1,2)$. Hence $\mathbf{G}$ is a connected absolutely almost simple group of type A_1, it is $\mathbb{R}$-isotropic, and its standard three-dimensional representation is $\mathbb{R}$-irreducible. The nonzero level $\mathscr{X}$ is Zariski closed. With

$$Q = \sqrt{3}\,q, \qquad N = 3\sqrt{2}\,n,$$

its real equation becomes

$$Q^2 + N^2 - a^2 = 2.$$

The parametrization

$$(a,\theta) \longmapsto (a, \sqrt{a^2+2}\cos\theta, \sqrt{a^2+2}\sin\theta), \qquad (a,\theta) \in \mathbb{R} \times (\mathbb{R}/2\pi\mathbb{Z}),$$

shows that $\mathscr{X}(\mathbb{R})$ is connected. Witt extension first produces an element $t \in \mathrm{O}(F)(\mathbb{R})$ taking any prescribed point of the level set to any other. If $\det t = -1$, choose an anisotropic vector in the orthogonal complement of the target point and compose t on the left with its reflection. That reflection fixes the target point and has determinant -1; hence the composite lies in $\mathrm{SO}(F)(\mathbb{R})$. Thus $\mathrm{SO}(F)(\mathbb{R})$ is transitive on the level set. The identity-component orbits have the full dimension of the level surface and are open; because their complements are unions of open orbits, each is also closed. Connectedness therefore leaves only one such orbit, proving $\mathscr{X}(\mathbb{R}) = \mathbf{G}(\mathbb{R})^\circ x_0$. The stabilizer of a nonisotropic vector is the fixed subgroup of the involution induced by reflection in that vector, so the orbit is affine symmetric.

The orthogonal complement of x_0 is

$$x_0^\perp = \{(3s,s,t) : s,t \in \mathbb{Q}\}, \qquad F(3s,s,t) = 6(s^2 - 3t^2). \tag{2.48}$$

Thus $\mathbf{H}$ is the norm-one torus of $K = \mathbb{Q}(\sqrt{3})$. Its character lattice is $\mathbb{Z}$, with the nontrivial element of $\mathrm{Gal}(K/\mathbb{Q})$ acting by -1; consequently it has no nonzero Galois-invariant character and no nontrivial $\mathbb{Q}$-character. The integral point x_0 supplies the required nonempty integral locus.

It remains to check the sector rather than merely the affine orbit. For the given representation and any Euclidean sphere in W, the Satake boundary is

$$\mathscr{X}^\infty = \{F = 0\} \cap S(W) = \mathcal{O}_+ \sqcup \mathcal{O}_-, \qquad \mathcal{O}_\pm := \{F = 0,\ \pm a > 0\} \cap S(W), \tag{2.49}$$

and $\mathbf{G}(\mathbb{R})^\circ$ is transitive on each displayed circle. For a nonzero null vector with $a, q, n \geq 0$,

$$a = \sqrt{3q^2 + 18n^2} > q - 4n.$$

Hence both the closure and the relative interior of $\mathcal{C}$ meet exactly $\mathcal{O}_+$, not $\mathcal{O}_-$. The intersection of $\partial\mathcal{C}$ with the Satake boundary consists of the two radial directions

$$(a,q,n) = (\sqrt{3},1,0), \qquad (a,q,n) = (3\sqrt{2},0,1). \tag{2.50}$$

These are exactly the two projective limits of the integral Pell families in Proposition 2.5. This two-point intersection has measure zero for the smooth measure class on the boundary orbit. This is precisely the admissibility and genericity condition required by the cited sector theorem. □

2.5 The 2-adic orbit sign and Borovoi's coefficient

Write $\widehat{\mathbb{Z}} = \prod_p \mathbb{Z}_p$. For the nonsquare ternary quadric under consideration, Borovoi associates to every local point an orbit sign $\nu_v \in \{\pm 1\}$, constant on $\mathrm{Spin}(F)(\mathbb{Q}_v)$-orbits and equal to $+1$

at almost every finite place. Thus $\nu_f((x_p)_p) = \prod_p \nu_p(x_p)$ is defined on $\mathscr{X}(\widehat{\mathbb{Z}})$. Because $\mathscr{X}(\mathbb{R})$ is connected, his normalization reads

$$c_{\mathscr{X}} = \frac{2m_f\{x \in \mathscr{X}(\widehat{\mathbb{Z}}) : \nu_f(x) = +1\}}{m_f(\mathscr{X}(\widehat{\mathbb{Z}}))}, \tag{2.51}$$

where m_f is the finite Tamagawa measure attached to the gauge form (2.37) and Borovoi's convergence factors.

Proposition 2.10 (Finite adelic sign balance)**.** *For the affine quadric $\mathscr{X} = \{F = -2\}$, the coefficient $c_{\mathscr{X}}$ in Borovoi's counting formula (Borovoi, 2001, Theorem 1.4.1 and Proposition 1.5.1) is exactly*

$$c_{\mathscr{X}} = 1. \tag{2.52}$$

Proof. Continue to write $A = \mathrm{diag}(1, -3, -18)$. Put $e = (1, 0, 0)$, $v = (1, 0, 1)$, and for every anisotropic vector w let

$$r_w(u) = u - \frac{2u^{\mathsf{T}}Aw}{w^{\mathsf{T}}Aw}\, w$$

be its orthogonal reflection. The element $g = r_v r_e$ is

$$g = \begin{pmatrix} -19/17 & 0 & -36/17 \\ 0 & 1 & 0 \\ -2/17 & 0 & -19/17 \end{pmatrix}. \tag{2.53}$$

Direct multiplication gives

$$g^{\mathsf{T}}Ag = A, \qquad \det g = 1. \tag{2.54}$$

Because $17 \in \mathbb{Z}_2^\times$, both g and g^{-1} have entries in $\mathbb{Z}_2$, so $g \in \mathrm{SO}(F, \mathbb{Z}_2)$. Its spinor norm is

$$\vartheta(g) = F(v)F(e) = -17 \quad \text{in } \mathbb{Q}_2^\times/\mathbb{Q}_2^{\times 2}. \tag{2.55}$$

We identify the stabilizer's spinor-norm image. By (2.48), over $\mathbb{Q}_2$ the orthogonal complement of x_0 is

$$F|_{x_0^\perp} = 6(s^2 - 3t^2) = 6N_{K_2/\mathbb{Q}_2}(s + t\sqrt{3}), \qquad K_2 = \mathbb{Q}_2(\sqrt{3}). \tag{2.56}$$

The odd unit 3 is not a square in $\mathbb{Q}_2$, so K_2 is a quadratic field and this binary plane is anisotropic. Put $u_0 = (3, 1, 0)$, so $F(u_0) = 6$. If $h \in \mathrm{SO}(F|_{x_0^\perp})(\mathbb{Q}_2)$, then hr_{u_0} is a two-dimensional orthogonal transformation of determinant -1, hence is a reflection r_w in some nonisotropic vector $w = (3s, s, t)$. Therefore $h = r_w r_{u_0}$ and

$$\vartheta(h) = F(w)F(u_0) = 36N_{K_2/\mathbb{Q}_2}(s + t\sqrt{3}) \mod \mathbb{Q}_2^{\times 2}.$$

Conversely these products realize every norm square class as w ranges through the nonzero vectors of the anisotropic plane. Hence

$$\vartheta\Big(\mathrm{Stab}_{\mathrm{SO}(F)}(x_0)(\mathbb{Q}_2)\Big) = N_{K_2/\mathbb{Q}_2}(K_2^\times) \mod \mathbb{Q}_2^{\times 2}. \tag{2.57}$$

For a quadratic extension, its norm group is the kernel of the Hilbert symbol with the defining square class (Voight, 2021, Corollary 13.4.6). The odd-unit formula (Voight, 2021, Equation 12.4.14) gives

$$(-17, 3)_2 = (-1)^{((-17-1)/2)((3-1)/2)} = -1. \tag{2.58}$$

Thus $-17 \notin N_{K_2/\mathbb{Q}_2}(K_2^\times)$.

By Witt extension, every $x \in \mathscr{X}(\mathbb{Q}_2)$ is tx_0 for some $t \in \mathrm{O}(F)(\mathbb{Q}_2)$. If $\det t = -1$, choose an anisotropic vector in $x^\perp$ and compose on the left with its reflection; this fixes x and changes the determinant. Hence every such x has the form $x = sx_0$ with $s \in \mathrm{SO}(F)(\mathbb{Q}_2)$. The image of $\mathrm{Spin}(F)(\mathbb{Q}_2)$ in $\mathrm{SO}(F)(\mathbb{Q}_2)$ is the kernel of the spinor norm, as used in (Borovoi, 2001, Lemma 3.1). Consequently the $\mathrm{Spin}(F)(\mathbb{Q}_2)$-orbit of $x = sx_0$ is determined by

$$\vartheta(s) \in \mathbb{Q}_2^\times / (\mathbb{Q}_2^{\times 2} N_{K_2/\mathbb{Q}_2}(K_2^\times)), \tag{2.59}$$

a quotient of order two. Equations (2.55) and (2.58) show that multiplication by g exchanges these two orbits. In Borovoi's convention, the orbit of x_0 has sign $\nu_2 = +1$ and the other orbit has sign $\nu_2 = -1$. Hence

$$\nu_2(gx) = -\nu_2(x) \qquad (x \in \mathscr{X}(\mathbb{Q}_2)). \tag{2.60}$$

Apply g at the prime 2 and the identity at every other finite prime. This gives a bijection of $\mathscr{X}(\widehat{\mathbb{Z}})$. It preserves the finite Tamagawa measure: at 2 this follows from invariance of the local measure, equivalently from $g^*F = F$ and $\det g = 1$, and at all other primes the map is the identity. Equation (2.60) shows that it exchanges the finite sign loci $\nu_f = +1$ and $\nu_f = -1$. They therefore have equal measure. Since $\mathscr{X}(\mathbb{R})$ is connected, Borovoi's formula becomes

$$c_{\mathscr{X}} = \frac{2m_f(\mathscr{X}(\widehat{\mathbb{Z}})_+)}{m_f(\mathscr{X}(\widehat{\mathbb{Z}}))} = 1.$$

The argument uses a measure-preserving, sign-reversing bijection; it does not require g to be an involution (indeed $g^2 \neq I$). □

2.6 Calibration and the counting constant

Theorem 2.11 (Calibrated reduced-root asymptotic)**.** *The number of reduced positive defect-one roots satisfies*

$$R(X) \sim C_0 X \qquad (X \to \infty), \tag{2.61}$$

where

$$\begin{aligned} C_0 &= \mathfrak{S}(\mathscr{X}) I_\infty \\ &= \frac{9}{2\pi^2} \log(2+\sqrt{3}) \log\left(\frac{3}{2\sqrt{2}}\right) \\ &= 0.035362047799731853\ldots. \end{aligned} \tag{2.62}$$

Proof. Let μ_{GOS} denote the invariant measure on $\mathscr{X}(\mathbb{R})$ with the normalization appearing in the Gorodnik–Oh–Shah sector theorem (Gorodnik et al., 2009, Theorem 1.6). The gauge measure m_∞ in (2.37) is also a nonzero $\mathbf{G}(\mathbb{R})^\circ$-invariant Radon measure. Because $\mathscr{X}(\mathbb{R})$ is a single homogeneous orbit, uniqueness of invariant measure gives

$$\mu_{\mathrm{GOS}} = \kappa m_\infty \tag{2.63}$$

for some $\kappa > 0$.

The scalar is fixed by applying both counting theorems to the same Euclidean balls and the full cone W. Gorodnik–Oh–Shah give

$$\#\{x \in \mathscr{X}(\mathbb{Z}) : |x|_2 < T\} \sim \kappa\, m_\infty\{x \in \mathscr{X}(\mathbb{R}) : |x|_2 < T\}.$$

Borovoi's Theorem 1.4.1 and Proposition 1.5.1, together with Proposition 2.10, give for the same balls

$$\#\{x \in \mathscr{X}(\mathbb{Z}) : |x|_2 < T\} \sim \mathfrak{S}(\mathscr{X})\, m_\infty\{x \in \mathscr{X}(\mathbb{R}) : |x|_2 < T\}.$$

The real volume here is asymptotic to a positive constant times T. Comparison therefore gives the invariant-measure calibration

$$\kappa = \mathfrak{S}(\mathscr{X}). \tag{2.64}$$

Lemma 2.8 and Lemma 2.9 now allow the sector theorem to be applied with the actual norm and chamber of the reduced-root problem:

$$\begin{aligned} R(X) &\sim \mu_{\mathrm{GOS}}\{u \in \mathscr{X}(\mathbb{R}) \cap \mathcal{C} : \|u\|_* \le X\} \\ &= \mathfrak{S}(\mathscr{X})\, m_\infty\{u \in \mathscr{X}(\mathbb{R}) \cap \mathcal{C} : 4a + 17n \le X\} \\ &\sim \mathfrak{S}(\mathscr{X}) I_\infty X. \end{aligned}$$

The switch between strict and weak inequalities is harmless by a $(1 \pm \varepsilon)X$ squeeze; on integral points it also follows directly from (2.47). Since $R(X) = R(\lfloor X \rfloor)$, the integer-height asymptotic extends to all real $X \to \infty$. Finally, substituting (2.28) and (2.40), and using $\zeta(2) = \pi^2/6$, gives exactly (2.62). □

2.7 Exact computation and the remaining open problem

The first five reduced roots are

$$(1, 4, 20), \quad (4, 4, 33), \quad (4, 8, 49), \quad (4, 15, 76), \quad (8, 24, 129). \tag{2.65}$$

The first root is the first admissible term of the $n = 0$ Pell family and the second is the first term of the $q = 0$ Pell family in Proposition 2.5. Exact integer enumeration in the ternary coordinates gives the following values. The residual column uses the full constant in (2.62), not a fitted constant.

X	$R(X)$	$R(X) - C_0 X$
10,000	353	−0.62048
20,000	715	7.75904
50,000	1,755	−13.10239
75,000	2,654	1.84642
100,000	3,535	−1.20478
125,000	4,460	39.74403
150,000	5,294	−10.30717
175,000	6,195	6.64164
200,000	7,109	36.59044

The enumeration loops over $n \ge 0$ and $a > 0$ with $4a + 17n \le X$, tests whether

$$q^2 = \frac{a^2 - 18n^2 + 2}{3}$$

is the square of a nonnegative integer satisfying $q < a + 4n$, and then reconstructs (x, y, z) with (2.15). Every retained triple is rechecked in the original polynomial (2.2) and against all ordering and wall inequalities. Separate exhaustive congruence enumerations agree with the formulae above at $p = 2, 3, 5, 7, 11, 13$ through the computed powers. These finite checks verify the implementation; the asymptotic theorem rests on the proofs above, not on numerical fitting.

The reproducibility script is

`supporting_materials/workbench/reproduction/t1_koide_exact/verify_koide_narain_root_density.py`.

Running it with `-max-z 200000` emits the coordinate, local-density, 2-adic isometry, and bounded-enumeration certificates recorded in

`supporting_materials/workbench/reproduction/t1_koide_exact/koide_verification_200000.json`.

The script uses vectorized square detection only to propose candidates; it rechecks every retained candidate with integer arithmetic in both coordinate systems and in the original polynomial Δ.

Open problem 2.12 (Square-root discrepancy)**. OPEN.** Determine whether, for every $\varepsilon > 0$,

$$R(X) - C_0 X = O_\varepsilon(X^{1/2+\varepsilon}). \tag{2.66}$$

The homogeneous-dynamics argument above and the exact effective-well-roundedness verification in theorem 9.2 below prove, for some fixed $\eta > 0$,

$$R(X) = C_0 X + O(X^{1-\eta}),$$

but do not prove (2.66). No unproved spectral gap, GRH assumption, or implicit "conditional theorem" is being used here: the sharper error term is an open problem in this workbench; see the primary-source comparison in section 14.1.

3 Exact ideal and unit-orbit parametrization of the unchanged sharp Koide count

Throughout this section the variables, equation, chamber, and height are exactly

$$a^2 - 3q^2 - 18n^2 = -2, \qquad a > 0, \quad q \geq 0, \quad n \geq 0, \quad a + 4n > q, \qquad 4a + 17n \leq X. \tag{3.1}$$

Every auxiliary symbol introduced below is connected to these same integer points by an explicitly proved map and inverse. No count for a normalized form, a rescaled lattice, or a substituted cutoff is used in place of (3.1). Thus

$$R(X) = \#\left\{(a, q, n) \in \mathbb{Z}^3 : \text{all conditions in (3.1) hold}\right\}. \tag{3.2}$$

The already calibrated leading constant is

$$C_0 = \frac{9}{2\pi^2} \log(2 + \sqrt{3}) \log\left(\frac{3}{2\sqrt{2}}\right). \tag{3.3}$$

The exact parametrization proved below has four pieces of data for each fixed $n \geq 1$: a positive integer C, a positive integer D, a square root E of 3 modulo D, and a nonnegative Pell exponent k. There is no quotient left implicit: equations (6.5), (6.6), and (6.8) construct (a, q, n) by finite integer operations, while (5.5) and the descent (6.4) recover (C, D, E, k) from (a, q, n). The face $n = 0$ is recorded separately and exactly.

This parametrization gives the exact sharp-cutoff formula (7.8). It also separates the discrepancy into the radial shifted-divisor sum and the unit-phase discrepancy in (7.15). The square-root estimate

$$R(X) - C_0 X = O_\varepsilon(X^{1/2+\varepsilon}) \tag{3.4}$$

is not proved or refuted here. What is proved unconditionally, by applying the effective symmetric-variety theorem to the exact chamber, is that there is a constant $\eta > 0$ for which

$$R(X) = C_0 X + O(X^{1-\eta}). \tag{3.5}$$

The cited general theorem does not supply $\eta \geq 1/2$, and its proof does not justify replacing (3.5) by (3.4).

4 Arithmetic of $\mathbb{Z}[\sqrt{3}]$

Put

$$K = \mathbb{Q}(\sqrt{3}), \qquad \mathcal{O} = \mathbb{Z}[\sqrt{3}], \qquad \epsilon = 2 + \sqrt{3}. \tag{4.1}$$

For $\alpha = u + v\sqrt{3} \in K$, write

$$\alpha' = u - v\sqrt{3}, \qquad \mathrm{N}(\alpha) = \alpha\alpha' = u^2 - 3v^2. \tag{4.2}$$

Lemma 4.1 (Ring, class number, and totally positive units)**.** *The ring of integers of K is $\mathcal{O}$. Every nonzero integral ideal of $\mathcal{O}$ is principal. Every unit has norm $+1$, and the totally positive unit group is*

$$\mathcal{O}^{\times,+} = \{\epsilon^j : j \in \mathbb{Z}\}. \tag{4.3}$$

Proof. Since $3 \equiv 3 \pmod 4$, the standard integral-basis calculation for a quadratic field gives $\mathcal{O}_K = \mathbb{Z}[\sqrt{3}]$ and discriminant 12. Minkowski's bound says that every ideal class in a real quadratic field of discriminant 12 has an integral representative of norm at most $\frac{2!}{2^2}\sqrt{12} = \sqrt{3} < 2$. Its norm must therefore be 1, so it is $\mathcal{O}$ and the ordinary class number is one.

The norm of a unit is ± 1. The equation $u^2 - 3v^2 = -1$ is impossible modulo 3, because it would give $u^2 \equiv 2 \pmod 3$. Hence every unit has norm $+1$. Dirichlet's unit theorem makes the positive units an infinite cyclic group. To identify its least generator, suppose $\gamma = u + v\sqrt{3} > 1$ and $\gamma' > 0$. Replacing γ by its inverse if necessary makes $v \geq 0$. If $v = 0$, then $\gamma = 1$. If $v \geq 1$, then $u^2 - 3v^2 = 1$ forces $u \geq 2$, and consequently $\gamma \geq 2 + \sqrt{3} = \epsilon$. Since ϵ itself has norm one, it is the least positive unit greater than one and generates the group. □

For $m \geq 1$ define

$$\chi_{12}(d) = \left(\frac{12}{d}\right), \qquad \rho(m) = \sum_{d \mid m} \chi_{12}(d). \tag{4.4}$$

Here the symbol is Kronecker's symbol; in particular

$$\chi_{12}(d) = \begin{cases} 0, & 2 \mid d \text{ or } 3 \mid d, \\ 1, & d \equiv 1, 11 \pmod{12}, \\ -1, & d \equiv 5, 7 \pmod{12}. \end{cases} \tag{4.5}$$

Lemma 4.2 (Ideal number)**.** *The number of integral ideals of $\mathcal{O}$ having absolute norm m is $\rho(m)$.*

Proof. The Dedekind zeta function factors as $\zeta_K(s) = \zeta(s)L(s, \chi_{12})$. Comparing Dirichlet coefficients gives the assertion. Equivalently, the equality follows prime by prime. At 2 and 3 there is one ramified prime ideal, so there is one ideal of norm p^e for every $e \geq 0$. If $\chi_{12}(p) = 1$, then p splits and there are $e + 1$ ideals of norm p^e. If $\chi_{12}(p) = -1$, then p is inert, so there is

one such ideal when e is even and none when e is odd. These are respectively the local values of $\sum_{j=0}^{e}\chi_{12}(p)^j$. Consequently, if $m=2^{e_2}3^{e_3}\prod_{p\ge5}p^{e_p}$, then

$$\rho(m)=\begin{cases}\displaystyle\prod_{\substack{p\mid m,\ p\ge5\\ p\equiv1,11\ (12)}}(e_p+1), & e_p \text{ is even for every } p\equiv5,7 \pmod{12},\\ 0, & \text{otherwise.}\end{cases}\tag{4.6}$$

□

5 Hermite-normal-form parameters for every ideal

For $m\ge1$ let

$$\mathcal{P}(m)=\left\{(C,D,E):\ \begin{array}{ll}C,D\in\mathbb{Z}_{>0}, & C^2D=m,\\ 0\le E<D, & E^2\equiv3 \pmod D\end{array}\right\}.\tag{5.1}$$

Proposition 5.1 (Exact HNF parametrization of ideals)**.** *The map*

$$(C,D,E)\longmapsto\mathfrak{a}(C,D,E):=C\big(D\mathbb{Z}+(E+\sqrt3)\mathbb{Z}\big)\tag{5.2}$$

is a bijection from $\mathcal{P}(m)$ to the integral ideals of $\mathcal{O}$ of norm m. Consequently

$$\#\mathcal{P}(m)=\sum_{C^2D=m}\#\{E \bmod D: E^2\equiv3 \pmod D\}=\rho(m).\tag{5.3}$$

Proof. Relative to the basis $1,\sqrt3$ of $\mathcal{O}$, the lattice in (5.2) has column basis

$$\begin{pmatrix}CD & CE\\ 0 & C\end{pmatrix}.$$

Its index is $C^2D=m$. It is stable under multiplication by $\sqrt3$ because

$$\sqrt3(CD)=D\,C(E+\sqrt3)-E(CD),$$
$$\sqrt3\,C(E+\sqrt3)=E\,C(E+\sqrt3)+\frac{3-E^2}{D}(CD),$$

and the last coefficient is integral. It is therefore an integral ideal of norm m.

Conversely, write the unique column Hermite normal form of an integral ideal $\mathfrak{a}$ as

$$\mathfrak{a}=A\mathbb{Z}+(B+C\sqrt3)\mathbb{Z},\qquad A,C>0,\quad 0\le B<A.$$

Stability of A under multiplication by $\sqrt3$ implies $C\mid A$ and $C\mid B$. Write $A=CD$ and $B=CE$. Stability of $B+C\sqrt3$ then implies $D\mid3-E^2$. The HNF inequality gives $0\le E<D$, and the index is $AC=C^2D$. HNF uniqueness proves injectivity and surjectivity.

For completeness, the last equality in (5.3) can be checked without zeta functions. Let $r_3(D)=\#\{E \bmod D: E^2\equiv3 \pmod D\}$. For $p\ge5$, Hensel's lemma gives $r_3(p^j)=2$ for every $j\ge1$ if $\chi_{12}(p)=1$, and $r_3(p^j)=0$ if $\chi_{12}(p)=-1$. Also $r_3(2)=r_3(3)=1$ and $r_3(2^j)=r_3(3^j)=0$ for $j\ge2$. Thus, for every prime power p^e,

$$\sum_{2c+j=e}r_3(p^j)=\sum_{j=0}^{e}\chi_{12}(p)^j.$$

Multiplicativity proves (5.3). □

The HNF parameters can be recovered directly from a generator. Suppose $\alpha = a + q\sqrt{3}$ has positive norm m, with $a > 0$ and $q \geq 0$. Put

$$C = \gcd(a, q), \qquad A = a/C, \qquad B = q/C, \qquad D = m/C^2. \tag{5.4}$$

If $D = 1$, put $E = 0$. If $D > 1$, then

$$\gcd(B, D) = 1, \qquad E \equiv AB^{-1} \pmod{D}, \qquad 0 \leq E < D. \tag{5.5}$$

Indeed, $\gcd(B, D) = \gcd(B, A^2 - 3B^2) = \gcd(B, A^2) = 1$, and $E^2 \equiv 3 \pmod{D}$. The principal ideal (α) is exactly $\mathfrak{a}(C, D, E)$: the element $A + B\sqrt{3}$ lies in $D\mathbb{Z} + (E + \sqrt{3})\mathbb{Z}$, so the principal ideal is contained in the HNF ideal, while both ideals have norm D after division by C.

6 Complete integral parametrization of the original equation

For $n \geq 1$ retain the exact integer

$$m_n = 18n^2 - 2. \tag{6.1}$$

It satisfies $m_n > 0$ and $m_n \equiv 1 \pmod{3}$.

Lemma 6.1 (Every ideal of norm m_n has positive-norm generators)**.** *If $\mathfrak{a}$ is an integral ideal of $\mathcal{O}$ of norm m_n, then every generator has norm $+m_n$, rather than $-m_n$.*

Proof. Lemma 4.1 makes $\mathfrak{a}$ principal. A generator $u + v\sqrt{3}$ has norm $\pm m_n$. Negative norm would give

$$u^2 - 3v^2 = -m_n,$$

and reduction modulo 3 would give $u^2 \equiv -m_n \equiv 2 \pmod{3}$, which is impossible. Thus the norm is $+m_n$. □

For each $\mathbf{p} = (C, D, E) \in \mathcal{P}(m_n)$, choose a generator of $\mathfrak{a}(C, D, E)$. By Lemma 6.1 its norm is m_n. Change its sign so that both real embeddings are positive, and multiply by a unique power of ϵ so as to obtain the unique generator

$$\alpha_{n,\mathbf{p}} = a_{n,\mathbf{p},0} + q_{n,\mathbf{p},0}\sqrt{3}, \qquad \sqrt{m_n} \leq \alpha_{n,\mathbf{p}} < \epsilon\sqrt{m_n}. \tag{6.2}$$

Existence and uniqueness follow from (4.3), because the half-open intervals $[\epsilon^j\sqrt{m_n}, \epsilon^{j+1}\sqrt{m_n})$ partition $\mathbb{R}_{>0}$. Since $\alpha'_{n,\mathbf{p}} = m_n/\alpha_{n,\mathbf{p}}$, the lower inequality in (6.2) implies $q_{n,\mathbf{p},0} \geq 0$.

Lemma 6.2 (Purely integral reduced seed and descent)**.** *For an integral pair (a, q) satisfying*

$$a^2 - 3q^2 = m_n, \qquad a > 0, \qquad q \geq 0,$$

the half-open inequalities

$$\sqrt{m_n} \leq a + q\sqrt{3} < \epsilon\sqrt{m_n}$$

are equivalent to the purely integral inequalities

$$a > 2q \geq 0. \tag{6.3}$$

If $a \leq 2q$, multiplication by $\epsilon^{-1} = 2 - \sqrt{3}$ is the integral inverse step

$$(a, q) \longmapsto (a^-, q^-) = (2a - 3q, 2q - a). \tag{6.4}$$

It preserves the norm, has $a^- > 0$ and $0 \leq q^- < q$, and can therefore be iterated until the unique pair satisfying (6.3) is reached.

Proof. Put $u = (a+q\sqrt{3})/\sqrt{m_n}$. Norm positivity and the sign conditions give $u \geq 1$. Moreover

$$\frac{a}{\sqrt{m_n}} = \frac{u+u^{-1}}{2}, \qquad \frac{q\sqrt{3}}{\sqrt{m_n}} = \frac{u-u^{-1}}{2}.$$

For $u \geq 1$, the ratio

$$\frac{a}{q} = \sqrt{3}\,\frac{u^2+1}{u^2-1}$$

(with value $+\infty$ when $q = 0$) is strictly decreasing. At $u = \epsilon$ it equals 2, since $\epsilon + \epsilon^{-1} = 4$. Thus $u < \epsilon$ is equivalent to $a > 2q$; the strict inequality exactly matches the half-open upper endpoint.

If $a \leq 2q$, norm positivity gives $a > \sqrt{3}q$, and hence

$$a^- = 2a - 3q > (2\sqrt{3}-3)q > 0, \qquad q^- = 2q - a \geq 0.$$

Also $a > \sqrt{3}q > q$ gives $q^- = 2q - a < q$. Thus the positive integer q strictly decreases at every inverse step, so the process terminates. A terminal pair has $a > 2q$. Conversely, for such a pair the second coordinate after one inverse step is negative, so no further inverse step stays in the nonnegative-q region. This proves uniqueness as well as termination. □

The preceding seed can be reconstructed from the HNF data by a bounded integer search; thus the parametrization does not require an unspecified choice of a generator. More precisely, for $\mathbf{p} = (C, D, E) \in \mathcal{P}(m_n)$ there is a unique pair (A, B) with

$$\begin{gathered} A, B \in \mathbb{Z}, \qquad A > 0, \qquad 0 \leq B, \qquad B^2 < D, \\ A^2 - 3B^2 = D, \qquad A \equiv EB \pmod{D}. \end{gathered} \tag{6.5}$$

It gives the generator in (6.2) by the exact integer formula

$$(a_{n,\mathbf{p},0}, q_{n,\mathbf{p},0}) = (CA, CB). \tag{6.6}$$

Indeed, existence follows by writing the reduced generator as $C(A+B\sqrt{3})$. The seed inequality $A > 2B$ gives $D = A^2 - 3B^2 > B^2$. Conversely, the conditions in (6.5) imply $A > 2B$ and put $A+B\sqrt{3}$ in $D\mathbb{Z}+(E+\sqrt{3})\mathbb{Z}$. Its principal ideal is contained in that HNF ideal, and both have norm D, so they are equal. Lemma 6.2 and HNF uniqueness give uniqueness. In particular one needs to test only the integers $0 \leq B \leq \lfloor\sqrt{D-1}\rfloor$ and the positive square $A = \sqrt{D+3B^2}$. The phase introduced below is therefore also explicit in the same integer data:

$$\tau_{n,(C,D,E)} = \frac{\log((A+B\sqrt{3})/\sqrt{D})}{\log\epsilon}. \tag{6.7}$$

Here the factor C cancels because $m_n = C^2D$; no ideal generator remains to be chosen.

For $k \geq 0$ define integers $a_{n,\mathbf{p},k}$ and $q_{n,\mathbf{p},k}$ by

$$a_{n,\mathbf{p},k} + q_{n,\mathbf{p},k}\sqrt{3} = \alpha_{n,\mathbf{p}}\epsilon^k. \tag{6.8}$$

Equivalently,

$$\begin{pmatrix} a_{n,\mathbf{p},k+1} \\ q_{n,\mathbf{p},k+1} \end{pmatrix} = \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} a_{n,\mathbf{p},k} \\ q_{n,\mathbf{p},k} \end{pmatrix}. \tag{6.9}$$

Theorem 6.3 (Integral parametrization, with inverse and multiplicity)**.** *The map*

$$(n, \mathbf{p}, k) \longmapsto (a_{n,\mathbf{p},k}, q_{n,\mathbf{p},k}, n), \qquad n \geq 1, \quad \mathbf{p} \in \mathcal{P}(m_n), \quad k \geq 0, \tag{6.10}$$

is a bijection onto all integral solutions of

$$a^2 - 3q^2 - 18n^2 = -2, \qquad a > 0,\ q \geq 0,\ n \geq 1. \tag{6.11}$$

Every such solution automatically satisfies $a + 4n > q$.

The inverse is as follows. From (a, q, n) *form* $\alpha = a + q\sqrt{3}$ *and recover* (C, D, E) *by* (5.4)–(5.5)*. There is a unique* $k \geq 0$ *such that*

$$\alpha = \alpha_{n,(C,D,E)}\epsilon^k. \tag{6.12}$$

Thus every point occurs with multiplicity exactly one.

Proof. Equation (6.8) preserves the norm, so its image satisfies

$$a_{n,\mathbf{p},k}^2 - 3q_{n,\mathbf{p},k}^2 = m_n,$$

which is exactly the equation in (6.11). Both embeddings of the generator are positive, hence $a > 0$; and $\alpha_{n,\mathbf{p}} \geq \sqrt{m_n}$ and $k \geq 0$ imply $q \geq 0$.

Conversely, a solution in (6.11) has $\mathrm{N}(a + q\sqrt{3}) = m_n > 0$. Since $a > 0$ and $q \geq 0$, the inequality $a^2 - 3q^2 = m_n > 0$ gives $a > \sqrt{3}q$; hence both embeddings are positive. Its principal ideal has unique HNF data (C, D, E). The quotient of $a + q\sqrt{3}$ by the canonical generator of that ideal is a totally positive unit, hence is ϵ^k for a unique $k \in \mathbb{Z}$. Lemma 6.2 supplies the completely integral recovery of k: iterate (6.4) precisely while $a \leq 2q$; the number of steps is k, and the terminal pair is $(a_{n,\mathbf{p},0}, q_{n,\mathbf{p},0})$. In particular $k \geq 0$. This proves surjectivity and the stated inverse. HNF uniqueness, seed uniqueness, and unit-exponent uniqueness prove multiplicity one.

It remains to prove the chamber wall. If a point with $n \geq 1$ had $q \geq a + 4n$, then $a \leq q - 4n$, and the unchanged equation would give

$$3q^2 + 18n^2 - 2 = a^2 \leq (q - 4n)^2.$$

After moving terms and dividing by 2, this is

$$q^2 + 4qn + n^2 \leq 1.$$

For nonnegative integers with $n \geq 1$, equality could occur only at $(q, n) = (0, 1)$, but that pair does not satisfy $q \geq a + 4n$ with $a > 0$. Thus the assumed wall inequality is impossible. □

The congruence needed by the original inverse map to the (x, y, z) coordinates is retained automatically: reducing $a^2 - 3q^2 - 18n^2 = -2$ modulo 2 gives

$$a \equiv q \pmod 2. \tag{6.13}$$

No parity class has been lost or added in Theorem 6.3.

Proposition 6.4 (The face $n = 0$)**.** *All points satisfying the original equation and chamber with* $n = 0$ *are*

$$a_k + q_k\sqrt{3} = (1 + \sqrt{3})\epsilon^k, \qquad k \geq 1. \tag{6.14}$$

They have height $4a_k$ *and occur with multiplicity one.*

Proof. When $n = 0$, the norm is -2. Its parity implies $a \equiv q \pmod 2$, and therefore

$$\frac{a + q\sqrt{3}}{1 + \sqrt{3}} = \frac{3q - a}{2} + \frac{a - q}{2}\sqrt{3} \in \mathcal{O}.$$

The quotient has norm one, so it is $\pm\epsilon^k$ by Lemma 4.1. The conditions $a > 0$ and $q \geq 0$ select the positive sign. At $k = 0$ one gets $(a, q) = (1, 1)$, which is excluded by the strict chamber wall $a > q$. At every $k \geq 1$, the recurrence (6.9) gives $a_{k+1} - q_{k+1} = a_k + q_k > 0$. For $k \leq -1$, direct use of $a = (t - 2/t)/2$ with $t = (1 + \sqrt{3})\epsilon^k \leq \sqrt{3} - 1 < \sqrt{2}$ gives $a < 0$. Hence precisely the exponents $k \geq 1$ occur. □

7 Exact divisor and sharp-height formulas

The number of unit-orbits at a fixed positive n has an especially simple divisor formula.

Proposition 7.1 (Orbit divisor sum and shifted factorization)**.** *For $n \geq 1$, the number of orbits under $(a, q) \mapsto (2a + 3q, a + 2q)$ among the solutions with $a > 0$ and $q \geq 0$ is*

$$\rho(m_n) = \sum_{d \mid 18n^2 - 2} \chi_{12}(d) = \rho(3n - 1)\rho(3n + 1). \tag{7.1}$$

Proof. The first equality is Theorem 6.3 and Proposition 5.1. For the second, write

$$m_n = 2(3n - 1)(3n + 1).$$

The two displayed linear factors have greatest common divisor dividing 2. The function $\rho = 1 * \chi_{12}$ is multiplicative and $\rho(2^j) = 1$ for all $j \geq 0$. Removing all powers of 2 therefore makes the two factors coprime without changing any of the three ρ values, proving the product identity. □

For $\mathbf{p} \in \mathcal{P}(m_n)$ define its exact unit phase by

$$\tau_{n,\mathbf{p}} = \frac{\log(\alpha_{n,\mathbf{p}}/\sqrt{m_n})}{\log \epsilon}, \qquad 0 \leq \tau_{n,\mathbf{p}} < 1. \tag{7.2}$$

For $n \geq 1$ put

$$Y_n(X) = \frac{X - 17n}{4}. \tag{7.3}$$

Whenever $Y_n(X) \geq \sqrt{m_n}$, put

$$L_n(X) = \frac{1}{\log \epsilon} \log\left(\frac{Y_n(X) + \sqrt{Y_n(X)^2 - m_n}}{\sqrt{m_n}} \right). \tag{7.4}$$

Let

$$\mathcal{N}(X) = \{n \in \mathbb{Z}_{\geq 1} : Y_n(X) \geq \sqrt{m_n}\}. \tag{7.5}$$

This is the exact finite range

$$X \geq 17n + 4\sqrt{18n^2 - 2}; \tag{7.6}$$

no asymptotic replacement of this boundary is made.

For the $n = 0$ face define

$$R_0(X) = \max\left\{ 0, \left\lfloor \frac{\log\left(\dfrac{X/4 + \sqrt{(X/4)^2 + 2}}{1 + \sqrt{3}} \right)}{\log \epsilon} \right\rfloor \right\}. \tag{7.7}$$

Theorem 7.2 (Exact sharp-height floor formula)**.** *For every real $X \geq 0$,*

$$R(X) = R_0(X) + \sum_{n \in \mathcal{N}(X)} \sum_{\mathbf{p} \in \mathcal{P}(m_n)} (\lfloor L_n(X) - \tau_{n,\mathbf{p}} \rfloor + 1)\,. \tag{7.8}$$

Every equality case in $4a + 17n \leq X$ is included by the floor in (7.8).

Proof. For a fixed positive n and parameter $\mathbf{p}$, put $t_k = \alpha_{n,\mathbf{p}} \epsilon^k$. Since $t_k \geq \sqrt{m_n}$,

$$a_{n,\mathbf{p},k} = \frac{1}{2}\left(t_k + \frac{m_n}{t_k}\right) \tag{7.9}$$

is nondecreasing, and is strictly increasing after the possible initial case $q = 0$. The unchanged height inequality is $a_{n,\mathbf{p},k} \leq Y_n(X)$. It has no solution unless $Y_n(X) \geq \sqrt{m_n}$. In that range, solving the quadratic inequality in t_k gives

$$t_k \leq Y_n(X) + \sqrt{Y_n(X)^2 - m_n}. \tag{7.10}$$

After division by $\sqrt{m_n}$ and logarithms, this is precisely

$$k + \tau_{n,\mathbf{p}} \leq L_n(X). \tag{7.11}$$

Because $L_n(X) \geq 0$ and $0 \leq \tau_{n,\mathbf{p}} < 1$, the number of integers $k \geq 0$ satisfying (7.11) is exactly $\lfloor L_n(X) - \tau_{n,\mathbf{p}} \rfloor + 1$; this expression is zero, not negative, when $L_n(X) < \tau_{n,\mathbf{p}}$. Equality is retained. Theorem 6.3 now proves the positive-n sum.

For $n = 0$, put $t = (1+\sqrt{3})\epsilon^k$. Since the norm is -2, $a_k = (t - 2/t)/2$. The inequality $4a_k \leq X$ is equivalent to

$$t \leq X/4 + \sqrt{(X/4)^2 + 2}.$$

Counting the integers $k \geq 1$ gives (7.7) and completes the proof. □

Write

$$J_n(X) = \lfloor L_n(X) \rfloor, \qquad \lambda_n(X) = L_n(X) - J_n(X) \in [0, 1), \tag{7.12}$$

and define the exact phase distribution

$$A_n(t) = \#\{\mathbf{p} \in \mathcal{P}(m_n) : \tau_{n,\mathbf{p}} \leq t\}, \qquad 0 \leq t < 1, \tag{7.13}$$

with discrepancy

$$\Delta_n(t) = A_n(t) - t\rho(m_n). \tag{7.14}$$

Since $\lfloor J + \lambda - \tau \rfloor + 1 = J + \mathbf{1}_{\{\tau \leq \lambda\}}$, Theorem 7.2 gives the exact identity

$$R(X) - C_0 X = R_0(X) + \left[\sum_{n \in \mathcal{N}(X)} \rho(m_n) L_n(X) - C_0 X\right] + \sum_{n \in \mathcal{N}(X)} \Delta_n(\lambda_n(X)). \tag{7.15}$$

This identity includes the two unbounded coordinate faces and every sharp height boundary point. Formula (7.7) also gives $R_0(X) = O(\log(2 + X))$, so the $n = 0$ face is below the requested $X^{1/2+\varepsilon}$ scale; the two positive-n brackets are the genuine analytic terms.

8 The phase term as exact Hecke-character data

The last sum in (7.15) is not determined by the divisor count $\rho(m_n)$. It records where the generator of each ideal lies inside a fundamental interval for the positive units.

For $\ell \in \mathbb{Z}$ and a nonzero integral ideal $\mathfrak{a} = (\gamma)$, define

$$\xi_\ell(\mathfrak{a}) = \exp\left(\frac{\pi i \ell}{\log \epsilon} \log\left|\frac{\gamma}{\gamma'}\right|\right). \tag{8.1}$$

This is independent of the generator: replacing γ by $\pm\epsilon^j\gamma$ changes the logarithm by $2j\log\epsilon$, and the exponential by $e^{2\pi i\ell j} = 1$. It is multiplicative, hence is the unramified ideal part of a unitary Hecke character of K. For $m = m_n$ and the canonical generator,

$$\xi_\ell(\mathfrak{a}(C,D,E)) = e^{2\pi i \ell \tau_{n,(C,D,E)}}. \tag{8.2}$$

Define its norm-m coefficient by

$$S_\ell(m) = \sum_{\mathrm{N}\,\mathfrak{a}=m} \xi_\ell(\mathfrak{a}). \tag{8.3}$$

Then $S_0(m) = \rho(m)$, whereas the nonzero $S_\ell(m_n)$ are the Fourier coefficients of the phase multiset.

The coefficient $S_\ell(m)$ also has a completely explicit Euler-factor description. It is multiplicative in m. If $p \geq 5$ is split, choose one prime ideal $\mathfrak{p}$ above p and put $\omega_{\ell,p} = \xi_\ell(\mathfrak{p})$. Since $\mathfrak{p}\overline{\mathfrak{p}} = (p)$ and $\xi_\ell((p)) = 1$, one has

$$S_\ell(p^e) = \sum_{j=0}^{e} \omega_{\ell,p}^{2j-e} \qquad (\chi_{12}(p) = 1). \tag{8.4}$$

If $p \geq 5$ is inert, then

$$S_\ell(p^e) = \begin{cases} 1, & 2 \mid e, \\ 0, & 2 \nmid e. \end{cases} \qquad (\chi_{12}(p) = -1). \tag{8.5}$$

Finally the unique primes above 2 and 3 are generated by $1+\sqrt{3}$ and $\sqrt{3}$, respectively. Since

$$\left|\frac{1+\sqrt{3}}{1-\sqrt{3}}\right| = \epsilon, \qquad \left|\frac{\sqrt{3}}{-\sqrt{3}}\right| = 1,$$

their factors are

$$S_\ell(2^e) = (-1)^{\ell e}, \qquad S_\ell(3^e) = 1. \tag{8.6}$$

These formulas enumerate every ideal factor with multiplicity one. At $\ell = 0$ they reduce prime by prime to the divisor sum $\rho(m)$; at $\ell \neq 0$ they show exactly what angular information the unweighted divisor sum omits.

They also retain the shifted factorization of m_n at every Fourier mode. Writing $3n-1 = 2^r u$ and $3n+1 = 2^s v$ with u, v odd, one has $\gcd(u,v) = 1$. Multiplicativity and (8.6) therefore give the exact identity

$$S_\ell(18n^2 - 2) = (-1)^\ell S_\ell(3n-1) S_\ell(3n+1). \tag{8.7}$$

Indeed, the exponent of 2 on the left is $1 + r + s$, while the product on the right has total exponent $r + s$; all odd prime factors are disjoint. Thus the angular obstruction is itself a shifted correlation, now of nontrivial Hecke coefficients rather than merely of their dimensions.

Here is an endpoint-exact Fourier statement. Suppose $0 < \lambda < 1$, and let

$$b_n(t) = \#\{\mathbf{p} \in \mathcal{P}(m_n) : \tau_{n,\mathbf{p}} = t\}. \tag{8.8}$$

Fejér summation of the Fourier series of $\mathbf{1}_{[0,\lambda]} - \lambda$ gives

$$\begin{aligned}\Delta_n(\lambda) = {} & \frac{1}{2}(b_n(0) + b_n(\lambda)) \\ & + \lim_{M\to\infty} \sum_{0<|\ell|\le M} \left(1 - \frac{|\ell|}{M+1}\right) \frac{1 - e^{-2\pi i\ell\lambda}}{2\pi i\ell} S_\ell(m_n).\end{aligned} \tag{8.9}$$

The two half-weights correct the midpoint convention at the circular endpoint 0 and at λ. When $\lambda = 0$, the exact value is simply $\Delta_n(0) = b_n(0)$.

Combining this with (8.7) makes the aggregate obstruction a literal shifted correlation. Define the exact endpoint contribution

$$\mathcal{E}(X) = \sum_{\substack{n\in\mathcal{N}(X)\\ \lambda_n(X)=0}} b_n(0) + \frac{1}{2} \sum_{\substack{n\in\mathcal{N}(X)\\ \lambda_n(X)>0}} \big(b_n(0) + b_n(\lambda_n(X))\big). \tag{8.10}$$

Since $\mathcal{N}(X)$ is finite, summation over n commutes with the Fejér limit and gives

$$\begin{aligned}\sum_{n\in\mathcal{N}(X)} \Delta_n(\lambda_n(X)) = {} & \mathcal{E}(X) \\ & + \lim_{M\to\infty} \sum_{0<|\ell|\le M} \left(1 - \frac{|\ell|}{M+1}\right) \frac{(-1)^\ell}{2\pi i\ell} \\ & \times \sum_{\substack{n\in\mathcal{N}(X)\\ \lambda_n(X)>0}} (1 - e^{-2\pi i\ell\lambda_n(X)}) S_\ell(3n-1) S_\ell(3n+1).\end{aligned} \tag{8.11}$$

Every coefficient and endpoint in this identity comes from the original weak height inequality; no smoothing has been inserted.

Equations (7.15) and (8.9) identify the analytic problem without discarding the original height. Its radial part is the sharply weighted shifted correlation

$$\sum_{n\in\mathcal{N}(X)} \rho(3n-1)\rho(3n+1)L_n(X), \tag{8.12}$$

and its angular part requires cancellation in the polynomially sampled Hecke coefficients

$$S_\ell(18n^2 - 2), \tag{8.13}$$

uniformly over the Fourier modes needed to approximate the discontinuous threshold $\tau \le \lambda_n(X)$. The endpoint (7.6) gives $L_n(X) = 0$ and has a square-root singularity in its derivative; it is not a fixed smooth weight. A bound for the unweighted orbit total $\rho(m_n)$ therefore does not bound the actual sharp-height count.

9 An unconditional effective exponent for the exact sector

This section records the power saving that is presently justified without assuming the square-root estimate.

Let

$$F(a,q,n)=a^2-3q^2-18n^2,\qquad \mathscr{X}=\{F=-2\},\qquad \mathbf{G}=\mathrm{Spin}(F). \tag{9.1}$$

Let $x_0=(1,1,0)$. Its orthogonal complement is

$$x_0^{\perp}=\{(3s,s,t):s,t\in\mathbb{Q}\},\qquad F(3s,s,t)=6(s^2-3t^2), \tag{9.2}$$

so the identity component of its stabilizer is the norm-one torus of $\mathbb{Q}(\sqrt{3})$. Its rational character group is zero because Galois conjugation acts by -1 on its geometric character lattice.

The form F is anisotropic over $\mathbb{Q}$. Indeed, suppose a primitive integral triple satisfies $F(a,q,n)=0$. Modulo 3 one first gets $3\mid a$. Writing $a=3a_1$ and dividing the equation by 3 gives $3a_1^2-q^2-6n^2=0$, hence $3\mid q$. Write $q=3q_1$ and divide by 3 again to get

$$a_1^2-3q_1^2-2n^2=0. \tag{9.3}$$

Modulo 3, the sum $a_1^2+n^2$ is zero, which forces $3\mid a_1$ and $3\mid n$. Thus 3 divides a,q,n, contradicting primitivity. In particular the associated arithmetic Fuchsian quotient is cocompact.

For $X>0$ let B_X be the exact subset of $\mathscr{X}(\mathbb{R})$ cut out by

$$a>0,\quad q\geq 0,\quad n\geq 0,\quad a+4n>q,\quad 4a+17n\leq X. \tag{9.4}$$

With the gauge measure

$$dm_\infty=\frac{dq\,dn}{2a}\qquad (a>0), \tag{9.5}$$

put

$$I_\infty=\frac{1}{2\sqrt{3}}\log\!\left(\frac{3}{2\sqrt{2}}\right). \tag{9.6}$$

Lemma 9.1 (Exact-sector volume with a power-saving remainder)**.** *As $X\to\infty$,*

$$m_\infty(B_X)=I_\infty X+O(1). \tag{9.7}$$

Moreover, the family B_X is effectively well-rounded in the sense of Benoist–Oh, Definition 13.1 (Benoist and Oh, 2012, p. 1931).

Proof. Introduce exact polar coordinates on the negative plane by

$$\sqrt{3}q=r\cos\theta,\qquad 3\sqrt{2}n=r\sin\theta,\qquad 0\leq\theta\leq\pi/2. \tag{9.8}$$

On $F=-2$ one has $r=\sqrt{a^2+2}$ and

$$dm_\infty=\frac{1}{6\sqrt{6}}\,da\,d\theta. \tag{9.9}$$

Write $b(\theta)=17\sin\theta/(3\sqrt{2})$. In the positive orthant the outer boundary is the unique nonnegative solution $A_X(\theta)$ of

$$4A_X(\theta)+b(\theta)\sqrt{A_X(\theta)^2+2}=X. \tag{9.10}$$

For all sufficiently large X, uniformly in θ,

$$A_X(\theta)=\frac{X}{4+b(\theta)}+O(X^{-1}). \tag{9.11}$$

To see this without a formal expansion, subtract $(4+b)A_X$ from (9.10) and use

$$\sqrt{A_X^2+2}-A_X=\frac{2}{\sqrt{A_X^2+2}+A_X}=O(X^{-1}),$$

where (9.10) uniformly gives $A_X \asymp X$. Integration of (9.11) in (9.9) gives

$$\frac{X}{6\sqrt{6}}\int_0^{\pi/2}\frac{d\theta}{4+\frac{17}{3\sqrt{2}}\sin\theta}+O(X^{-1})=I_\infty X+O(X^{-1}).$$

For the asserted exact value, set $c=17/(3\sqrt{2})$, $d=\sqrt{c^2-16}=1/(3\sqrt{2})$, and $t=\tan(\theta/2)$. Then

$$\begin{aligned}\int_0^{\pi/2}\frac{d\theta}{4+c\sin\theta}&=\int_0^1\frac{dt}{2t^2+ct+2}\\&=\frac{1}{d}\left[\log\left(\frac{4t+c-d}{4t+c+d}\right)\right]_{t=0}^{t=1}\\&=3\sqrt{2}\log\left(\frac{3}{2\sqrt{2}}\right).\end{aligned}$$

Multiplication by $1/(6\sqrt{6})$ is exactly the constant in (9.6). If the strict chamber wall fails, then the calculation in the proof of Theorem 6.3, now over $\mathbb{R}$, gives $q^2+4qn+n^2\le 1$. The removed real subset is therefore compact. For all large X it contributes a fixed finite measure, proving (9.7).

It remains to check effective well-roundedness for the actual sector. Let U_δ be the image in the fixed vector representation of the group ball used by Benoist–Oh (Benoist and Oh, 2012, Definition 13.1). For every linear functional ℓ on $\mathbb{R}^3$ there is a constant, independent of x and δ, such that

$$|\ell(xg)-\ell(x)|\le C_\ell\delta\|x\| \qquad (g\in U_\delta). \tag{9.12}$$

Apply this to the five literal boundary functionals

$$a, \qquad q, \qquad n, \qquad a+4n-q, \qquad 4a+17n.$$

On the unbounded part of the positive sheet, $\|x\|\asymp a+1$, and the last functional is also comparable to $\|x\|$. Thus membership in the outer height inequality can change only in a shell of radial width $O(\delta X)$. Membership in $q\ge 0$ or $n\ge 0$ can change only in angular strips of width $O(\delta)$ about $\theta=\pi/2$ or $\theta=0$. The boundary $a=0$ is compact. The boundary $a+4n-q=0$ is compact by the calculation preceding (9.7); moreover its restriction to $\mathscr{X}$ is regular, because proportionality of $(1,-1,4)$ to $(2a,-6q,-36n)$ is incompatible with $a,q,n\ge 0$. Consequently its group-action boundary tube has measure $O(\delta)$. By (9.9), all these regions together have measure $O(\delta X)$. The same analysis applies to the inner intersection that defines $B^-_{X,\delta}$. Hence

$$m_\infty(B^+_{X,\delta}\setminus B^-_{X,\delta})\ll \delta\, m_\infty(B_X), \tag{9.13}$$

which is precisely Definition 13.1(2), with boundary exponent 1, for the group-action thickenings rather than merely Euclidean tubes.

For Definition 13.1(3) of Benoist and Oh (2012), take the archimedean height to be the Euclidean norm in the displayed integral basis. It is bounded below on $\mathscr{X}$. Uniformly for $0<\delta<1$, the thickening $B^+_{X,\delta}$ lies in a fixed finite enlargement of the polar sector; there

$H_\infty(x) \asymp 1+a$ and the measure in a radial interval is $O(da)$. Hence for every $s>0$, uniformly in δ,

$$\int_{B_X^+} \|u\|^{-s}\, dm_\infty(u) \ll \begin{cases} X^{1-s}, & 0<s<1,\\ \log X, & s=1,\\ 1, & s>1. \end{cases}$$

Each line is $O(m_\infty(B_X)^{1-\delta_s})$, for example with $\delta_s=\min(s/2,1/2)>0$. This is the remaining condition. Since $S=\{\infty\}$, Definition 13.1(1), concerning a finite-place compact open subgroup, is void. The choices of strict or weak inequality on the finitely many piecewise smooth boundary arcs do not change these neighborhood estimates. □

Theorem 9.2 (Unconditional effective conclusion)**.** *There exists $\eta>0$, depending only on the fixed form, lattice, chamber, and height in (3.1), such that*

$$R(X) = C_0 X + O(X^{1-\eta}). \tag{9.14}$$

Proof. Use the standard vector representation $\rho:\mathrm{Spin}(F)\to \mathrm{SO}(F)\subset \mathrm{GL}_3$, whose kernel is finite, and the displayed integral basis of the exact lattice $\mathbb{Z}^3$. Let

$$\mathbf{G}_{\mathbb{Z}} = \{g\in \mathbf{G}(\mathbb{Q}) : \rho(g)\mathbb{Z}^3=\mathbb{Z}^3\}.$$

This is the integral group attached to the chosen representation and lattice in the Benoist–Oh setup (Benoist and Oh, 2012, pp. 1892–1893); take their congruence subgroup Γ_S to be the level-one group $\mathbf{G}_{\mathbb{Z}}$ itself. The group $\mathbf{G}$ is connected, simply connected, and almost $\mathbb{Q}$-simple. It is noncompact over $\mathbb{R}$ because F has signature $(1,2)$. Thus $S=\{\infty\}$ contains every noncompact archimedean place and G_S is noncompact. The orbit $x_0\mathbf{G}$ is the affine symmetric variety $\mathscr{X}$, and its stabilizer has no nontrivial rational character by (9.2). These verify every standing hypothesis on pp. 1892–1893 of Benoist and Oh (2012). Their Theorem 12.5 supplies the required effective equidistribution of the symmetric-stabilizer translates.

The Borel–Harish-Chandra finiteness theorem quoted in (Benoist and Oh, 2012, p. 1892) p. 1892 says that $\mathscr{X}(\mathbb{Z})$ is a finite union of $\mathbf{G}_{\mathbb{Z}}$-orbits. A rational representative of any such orbit again has a symmetric stabilizer with no rational character: its orthogonal complement is anisotropic over $\mathbb{Q}$, since an isotropic vector in it would be an isotropic vector for F. Apply Benoist–Oh, Theorem 13.2 (Benoist and Oh, 2012, pp. 1931–1932), to each orbit and to the exact effectively well-rounded family B_X of Lemma 9.1. For each orbit it gives its invariant-volume main term with error $O(m_\infty(B_X)^{1-\eta_j})$ for some $\eta_j>0$. There are finitely many orbits, so their sum has the same form with $\eta=\min_j \eta_j>0$.

All invariant measures on the same real homogeneous variety are proportional. Hence the sum of the finitely many orbit main terms is $K\, m_\infty(B_X)$ for one fixed $K>0$. Lemma 9.1 and the already established asymptotic $R(X)\sim C_0X$ force $KI_\infty = C_0$. Therefore

$$K\, m_\infty(B_X) = K(I_\infty X + O(1)) = C_0X + O(1).$$

Since $m_\infty(B_X)\asymp X$, summing the effective errors proves (9.14); the $O(1)$ term is absorbed after decreasing η if necessary. The theorem is applied first to the sequence of positive integral X. For real X, the original height $4a+17n$ is an integer, so $R(X)=R(\lfloor X\rfloor)$; replacing X by $\lfloor X\rfloor$ changes C_0X by $O(1)$ and proves the same estimate. □

Remark 9.3 (What exponent has actually been obtained). Theorem 9.2 is an unconditional exponent strictly smaller than 1: the error is $O(X^{1-\eta})$ for some $\eta>0$. Neither the statement nor the proof of (Benoist and Oh, 2012, Theorem 13.2) identifies an exponent $1-\eta\le 1/2$ for this sharp sector. Tracing their Sobolev and mixing constants would make a positive η effective in principle, but that calculation has not been carried out in this section and cannot be replaced by the assertion $\eta=1/2$.

10 Exact obstruction to the square-root claim

There are two complementary descriptions of the remaining obstruction; the first is an exact identity, while the second explains the relevant spectral mechanisms without asserting an unproved intertwining formula.

First, the elementary ideal description gives the literal identity (7.15). Thus the square-root claim is equivalent to a bound of that size for the displayed sum of the radial bracket, the aggregate phase discrepancy, and $R_0(X)$. Controlling the radial and phase terms separately is a sufficient route, though the exact identity also permits cancellation between them. Formula (8.9) shows that a separate estimate of the latter entails uniform cancellation in the nonzero archimedean Hecke modes (8.13). Merely using $\#\mathcal{P}(m_n) = \rho(m_n)$, or replacing each phase distribution by Lebesgue measure, deletes a term that is exactly present in $R(X)$.

Second, F is rationally anisotropic, so $\mathbf{G}$ gives a cocompact arithmetic Fuchsian quotient. A negative-norm vector represents an oriented geodesic, and its stabilizer is the split real torus appearing in (9.2). In the standard radial torus-orbit counting kernel, the spectral expansion has two possible sources larger than the square-root scale:

1. an exceptional eigenvalue $\lambda = s(1-s) < 1/4$ can contribute $c_s X^s$ with $s > 1/2$, unless the exact torus period and exact sector transform make $c_s = 0$;
2. after exceptional terms are absent or removed, estimating the tempered spectral sum for a discontinuous cutoff at the conjectural $X^{1/2+\varepsilon}$ scale requires cancellation beyond the general effective-mixing argument.

For the present nonradial chamber, one must first construct a remainder-preserving sector spectral expansion before these candidate terms can even be asserted with their exact coefficients. An actual refutation by a secondary term would then require proving both $s > 1/2$ and $c_s \neq 0$ for this fixed arithmetic quotient and exact chamber. No such construction or nonvanishing has been established here. Conversely, absence of exceptional terms alone would not estimate the full tempered sharp-cutoff sum. This is why the exact parametrization neither proves nor refutes (3.4), while it does isolate the data a future proof must bound.

For comparison only, not as a theorem about the present nonradial sector, the classical hyperbolic lattice-point problem for a conjugacy class has an unconditional $O(X^{2/3})$ remainder due to Good (Good, 1983, Theorem 4), with the modern proof of Chatzakos–Petridis (Chatzakos and Petridis, 2016, Theorem 1.1 and Conjecture 5.7), and a conjectural $O_\varepsilon(X^{1/2+\varepsilon})$ scale. That published $O(X^{2/3})$ theorem counts a different radial kernel. An exact intertwining of its kernel with the simultaneous chamber and height in (3.1) has not been proved, so $O(X^{2/3})$ is not claimed here for $R(X)$.

The independent finite replay used to check the HNF multiplicities, seed reduction, inverse descent, and divisor factorization for $1 \leq n \leq 200$ is retained at `supporting_materials/workbench/evidence/koide_exact_checkpoint_20260904/parametrization/verify_parametrization.py`; its SHA–256 digest is `061c5e22258628169fd1ecb8bcc5e7cf8b8d19bfdb918650668c3e90f2f9730d`. That replay is a finite consistency check. The proofs above, rather than the finite range, establish the parametrization for every admissible integer.

11 Boundary ledger

For clarity, every boundary used above has the following exact treatment.

- $a > 0$ is strict. There is no omitted integral solution with $a = 0$, because the original equation would give a multiple of 3 equal to 2.
- $q = 0$ is included. It corresponds to phase $\tau = 0$, is included by $\tau \le \lambda$, and is retained in the endpoint term $b_n(0)/2$ in (8.9).
- $n = 0$ is included and is counted exactly by $R_0(X)$; the sole solution $(a, q, n) = (1, 1, 0)$ on $a + 4n = q$ is excluded.
- $a + 4n = q$ is strict. Its real intersection is bounded and its sole integral point in the nonnegative chamber is $(1, 1, 0)$.
- $4a + 17n = X$ is included. Equality is included in (7.10), (7.11), and every floor in (7.8).
- The canonical unit interval is half-open, $[\sqrt{m_n}, \epsilon\sqrt{m_n})$. Consequently no generator is counted twice at a unit boundary.
- Conjugate ideals are distinct unless the ideal is fixed by conjugation. They yield distinct canonical phases and distinct $q \ge 0$ unit-orbits; no division by 2 is made. When $q = 0$, the ideal is self-conjugate and still occurs once by HNF uniqueness.

12 Exact unit-orbit census through height 10^8

The census retains exactly the same equation, chamber, strict wall, and height as the preceding theorems:

$$(a, q, n) \in \mathbb{Z}^3, \qquad a^2 - 3q^2 - 18n^2 = -2, \qquad a > 0, \quad q, n \ge 0, \quad a + 4n > q, \qquad 4a + 17n \le X. \tag{12.1}$$

It does not infer an asymptotic from finite data. Its exact fixed-n equation and factorization are

$$a^2 - 3q^2 = m_n, \qquad m_n = 18n^2 - 2 = 2(3n - 1)(3n + 1). \tag{12.2}$$

For each $n \ge 1$, every ideal of norm m_n is enumerated by its Hermite-normal-form triple

$$(C, D, E) \in \mathbb{Z}^2_{>0} \times \mathbb{Z}, \qquad C^2 D = m_n, \qquad 0 \le E < D, \qquad E^2 \equiv 3 \pmod{D}. \tag{12.3}$$

The generator is reduced by exact multiplication by the fundamental unit $2 + \sqrt{3}$ to the unique seed $a_0 > 2q_0 \ge 0$. Its forward unit orbit is

$$\begin{pmatrix} a_{k+1} \\ q_{k+1} \end{pmatrix} = \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} a_k \\ q_k \end{pmatrix}, \qquad k \ge 0. \tag{12.4}$$

Each emitted point is reconstructed by

$$x = \frac{a + 4n - q}{2}, \qquad y = \frac{a + 4n + q}{2}, \qquad z = 4a + 17n, \tag{12.5}$$

and is rechecked in both coordinate systems and in the original polynomial $\Delta(x, y, z) = -1$. Thus the accelerated enumeration does not replace the original objects by a floating-point or rescaled surrogate.

12.1 Complete result and checkpoint table

The complete integer-domain audit for $1 \le X \le 100{,}000{,}000$ gives

$$\boxed{R(100{,}000{,}000) = 3{,}536{,}002.} \tag{12.6}$$

The endpoint in n is 2,943,725; the enumeration contains 2,327,513 canonical unit orbits. The following checkpoints are exact integer counts. The residuals use the proved constant C_0 in (2.62).

X	$R(X)$	$R(X) - C_0X$	$(R(X) - C_0X)/\sqrt{X}$
200,000	7,109	36.590440053629	0.081818711287
250,000	8,854	13.488050067037	0.026976100134
500,000	17,695	13.976100134073	0.019765190359
1,000,000	35,381	18.952200268147	0.018952200268
2,000,000	70,637	−87.095599463706	−0.061585888992
5,000,000	176,943	132.761001340734	0.059372524752
10,000,000	353,598	−22.477997318532	−0.007108166877
20,000,000	707,458	217.044005362937	0.048532515010
50,000,000	1,768,230	127.610013407342	0.018046781166
100,000,000	3,536,002	−202.779973185317	−0.020277997319

12.2 Certified extrema over every integer

Write $E(X) = R(X) - C_0X$. Between consecutive occupied heights, $R(X)$ is constant and both $E(X)$ and $E(X)/\sqrt{X}$ are strictly decreasing, since

$$\frac{d}{dX}\left(\frac{R}{\sqrt{X}} - C_0\sqrt{X}\right) = -\frac{R}{2X^{3/2}} - \frac{C_0}{2\sqrt{X}} < 0. \tag{12.7}$$

Checking both integer endpoints of every constant-count interval therefore certifies extrema over every integer in the audited range, rather than at selected checkpoints. On $1 \le X \le 10^8$, the maximum signed residual is

$$\begin{aligned} E(77{,}078{,}404) &= 721.79342495712975696566815535386910933 79\ldots, \\ R(77{,}078{,}404) &= 2{,}726{,}372, \end{aligned} \tag{12.8}$$

and the minimum signed, hence largest absolute, residual is

$$\begin{aligned} E(32{,}461{,}294) &= -812.83006914880688970919660067974916559 33\ldots, \\ R(32{,}461{,}294) &= 1{,}147{,}085. \end{aligned} \tag{12.9}$$

The maximum of $|E(X)|/\sqrt{X}$ over the full audited range is attained at $X = 160$, where

$$\frac{|E(160)|}{\sqrt{160}} = 0.18515707693408214138401018551531723381 6\ldots.$$

On $10{,}000 \le X \le 10^8$, the maximum is

$$\begin{aligned} \frac{|E(867{,}240)|}{\sqrt{867{,}240}} &= 0.17140019280881455972894965803234555250 2\ldots, \\ R(867{,}240) &= 30{,}827, \end{aligned} \tag{12.10}$$

and the root entering at that height is

$$(a, q, n) = (199997, 115061, 3956), \qquad (x, y, z) = (50380, 165441, 867240). \tag{12.11}$$

The bound $|E(X)| \le 0.171\sqrt{X}$ first fails on this tail at $X = 867{,}216$, where $R(X) = 30{,}826$ and

$$E(X) = 159.466355307741222033044935323431981719 9\ldots.$$

No counterexample to the finite statement with constant 0.172 occurs through 10^8. These are bounded computations, not a claim that either constant persists.

The audit finds 12,916 sign changes. The 3,536,002 roots occupy 2,472,015 distinct heights; the maximum shell multiplicity is 35, at $z = 92{,}923{,}120$, and the largest gap between consecutive occupied heights is 564, from 71,586,276 to 71,586,840.

12.3 Independent replay and exact algorithm

An independent pass decompressed and rechecked all 3,536,002 six-coordinate rows, recomputed every checkpoint, verified the coordinate maps, equations, ordering, chamber, height, nonduplication, and gzip CRC, and returned `PASS`. It also compared the literal (n, a) scan through $X = 20{,}000$, every generalized-Pell seed for $1 \le n \le 1000$, and the full six-coordinate point set through $X = 200{,}000$.

The reproducibility sources and receipts are retained project-relatively under `supporting_materials/workbench/evidence/koide_exact_checkpoint_20260904/census`. The two 10^8-range outputs and the full coordinate receipt are in its `final_100000000` subdirectory. The principal digests are

artifact	SHA-256
`koide_fast_enum.py`	`dddb73b828b63be08e2ff0b5346edb32 31662005f83d5b2448c06fef6ba31017`
`koide_discrepancy_100000000.json`	`da92f0866074e530b8174a48915caa0f 65cd4cd8957e209306adc200739d1961`
`koide_roots_through_ 100000000.csv.gz`	`dfc313492aedb396dcfa10520bfefc699 ba601357bc9933d921dfa3884310bd4`
`coordinate_audit_receipt.json`	`3010363108e7b460b7331a92f3263bbd 77992aedfc3064973750e16d707da80f`

The exact algorithm first finds the largest admissible n by integer binary search, then factors the unchanged integers $3n - 1$ and $3n + 1$ with a smallest-prime-factor sieve. It enumerates every HNF triple in (12.3), constructs all roots of 3 modulo D by prime-power lifting and the Chinese remainder theorem, reduces each generator to the unique seed, and iterates (12.4) until the literal height fails. For each n, the independently checked identity

$$\#\mathcal{P}(m_n) = \rho(m_n) = \rho(3n - 1)\rho(3n + 1)$$

certifies the orbit multiplicity. The work is

$$X^{1+o(1)} + O(R(X))$$

integer operations before the optional height sort. This complexity and the finite census use no conjectural discrepancy estimate.

13 Exact quaternion bridge: order, shell, and hyperbolic conjugacy classes

The proposed quaternionic realization is correct. The displayed lattice is an order in an indefinite rational quaternion division algebra of discriminant 6. The parity-restricted ternary shell is exactly the reduced-trace-4, reduced-norm-1 locus in that order. Every resulting element is primitive hyperbolic, with eigenvalues $2 \pm \sqrt{3}$ and translation length $2\log(2+\sqrt{3})$.

The centralizer statement is correct when written intrinsically as

$$\mathcal{O} \cap \mathbb{Q}(\gamma) = \mathbb{Z}[\gamma - 2] \cong \mathbb{Z}[\sqrt{3}].$$

The entire integral shell is a finite union of conjugacy classes under $\Gamma = \mathcal{O}^1$. This gives an exact reduction of the Koide count to a finite sum of sharp, non-radial sector counts inside primitive hyperbolic conjugacy classes of a cocompact arithmetic Fuchsian group.

It does **not** prove the open estimate

$$R(X) - C_0 X = O_\varepsilon(X^{1/2+\varepsilon}).$$

Rather, it converts the count exactly into a sharp sector problem on a finite union of hyperbolic conjugacy classes. This places it beside, but does not prove it equivalent to, the standard hyperbolic conjugacy-class lattice-point problem. Even for the radial Huber cutoff, the general pointwise theorem gives an $O(X^{2/3})$ remainder after all exceptional spectral terms are included, while the $O(X^{1/2+\varepsilon})$ remainder is conjectural. The Koide cutoff is a different sharp polyhedral-sector height, so the standard radial theorem cannot be imported without a new transfer or trace-formula argument.

13.1 Quaternion algebra and reduced invariants

Let

$$B = (18, -6)_{\mathbb{Q}} = \mathbb{Q}\langle I, J\rangle/(I^2 - 18,\ J^2 + 6,\ IJ + JI), \qquad K = IJ.$$

Then

$$K^2 = -I^2 J^2 = 108.$$

For

$$x = x_0 + x_1 I + x_2 J + x_3 K,$$

the standard involution, reduced trace, and reduced norm are

$$\bar{x} = x_0 - x_1 I - x_2 J - x_3 K, \qquad \mathrm{trd}(x) = x + \bar{x} = 2x_0,$$

and

$$\mathrm{nrd}(x) = x\bar{x} = x_0^2 - 18x_1^2 + 6x_2^2 - 108x_3^2. \tag{13.1}$$

All signs in (13.1) are forced by $\mathrm{nrd}(I) = -18$, $\mathrm{nrd}(J) = 6$, and $\mathrm{nrd}(K) = -108$.

13.2 The proposed lattice is an order

Put

$$E = \frac{I+J}{2}, \qquad F_0 = \frac{I-J}{2}, \qquad G = \frac{K}{2},$$

and let

$$\mathcal{O} = \mathbb{Z}1 \oplus \mathbb{Z}E \oplus \mathbb{Z}F_0 \oplus \mathbb{Z}G. \tag{13.2}$$

The displayed elements are a rational basis of B, because

$$I = E + F_0, \qquad J = E - F_0, \qquad K = 2G.$$

Direct multiplication gives the complete nontrivial table

	E	F_0	G
E	3	$6 - G$	$6E - 3F_0$
F_0	$6 + G$	3	$3E - 6F_0$
G	$-6E + 3F_0$	$-3E + 6F_0$	$27.$

(13.3)

Every structure constant in (13.3) is integral. Thus (13.2) is a unital subring and a full rank-four $\mathbb{Z}$-lattice in B, hence an order. In particular, the unital ring sometimes denoted $\mathbb{Z}[1, E, F_0, G]$ is exactly the direct sum in (13.2).

An arbitrary order element is

$$x = r + sE + tF_0 + uG = r + \frac{s+t}{2}I + \frac{s-t}{2}J + \frac{u}{2}K \qquad (r, s, t, u \in \mathbb{Z}), \tag{13.4}$$

and (13.1) gives

$$\mathrm{trd}(x) = 2r, \qquad \mathrm{nrd}(x) = r^2 - 3s^2 - 12st - 3t^2 - 27u^2. \tag{13.5}$$

The reduced-trace pairing on the basis $(1, E, F_0, G)$ is

$$\big(\mathrm{trd}(b_i b_j)\big)_{i,j} = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 6 & 12 & 0 \\ 0 & 12 & 6 & 0 \\ 0 & 0 & 0 & 54 \end{pmatrix},$$

with determinant

$$-11664 = -108^2. \tag{13.6}$$

Thus the reduced discriminant of this order is 108, not 6. The algebra has discriminant 6, proved below, so $\mathcal{O}$ is nonmaximal and has index 18 in a maximal order containing it. This

index is not an Eichler level: the order is nonmaximal at the ramified primes 2 and 3, so it is not an Eichler order in the standard sense. Any optimal-embedding or trace-formula result that assumes a maximal order must account for this exact nonmaximal order instead of replacing $\mathcal{O}$.

13.3 Trace-zero lattice and exact shell bijection

Equation (13.5) gives

$$\mathcal{O}^0 := \{x \in \mathcal{O} : \mathrm{trd}(x) = 0\} = \mathbb{Z}E \oplus \mathbb{Z}F_0 \oplus \mathbb{Z}G.$$

Writing

$$q = s + t, \qquad a = s - t, \qquad n = u$$

gives the exact description

$$\boxed{\mathcal{O}^0 = \left\{ \frac{qI + aJ + nK}{2} : a, q, n \in \mathbb{Z},\ a \equiv q \pmod 2 \right\}.} \tag{13.7}$$

The inverse formulas are

$$s = \frac{q+a}{2}, \qquad t = \frac{q-a}{2}, \qquad u = n,$$

which are integral exactly when $a \equiv q \pmod 2$.

Define

$$F(a,q,n) = a^2 - 3q^2 - 18n^2, \qquad v(a,q,n) = \frac{qI + aJ + nK}{2},$$

and

$$\gamma(a,q,n) = 2 + v(a,q,n). \tag{13.8}$$

Distinct pure quaternion basis elements anticommute, so

$$v(a,q,n)^2 = \frac{18q^2 - 6a^2 + 108n^2}{4} = -\frac{3}{2}F(a,q,n), \tag{13.9}$$

and

$$\mathrm{nrd}(v) = \frac{3}{2}F(a,q,n).$$

Since v has reduced trace zero,

$$\mathrm{trd}(\gamma) = 4, \qquad \mathrm{nrd}(\gamma) = 4 + \mathrm{nrd}(v) = 4 + \frac{3}{2}F(a,q,n). \tag{13.10}$$

Therefore (13.8) defines a bijection

$$\boxed{\left\{(a,q,n)\in\mathbb{Z}^3 : F(a,q,n)=-2,\ a\equiv q \pmod 2\right\} \xrightarrow{\sim} \{\gamma\in\mathcal{O} : \mathrm{trd}(\gamma)=4,\ \mathrm{nrd}(\gamma)=1\}.} \tag{13.11}$$

The inverse sends

$$\gamma = 2 + sE + tF_0 + uG$$

to

$$(a,q,n) = (s-t, s+t, u).$$

There is no hidden congruence loss. On the integral equation $F=-2$, reduction modulo 2 gives

$$a^2 - q^2 \equiv 0 \pmod 2,$$

so $a\equiv q \pmod 2$ is automatic. The parity condition in (13.11) records the order lattice exactly but removes no integral shell point.

Finally,

$$(\gamma-2)^2 = 3, \qquad \gamma^2 - 4\gamma + 1 = 0, \qquad \gamma^{-1} = 4-\gamma = 2 - v \in \mathcal{O}. \tag{13.12}$$

Thus every element on the right of (13.11) lies in $\Gamma = \mathcal{O}^1$.

13.4 Division over $\mathbb{Q}$ and splitting over $\mathbb{R}$

Square factors may be removed from a Hilbert-symbol presentation, so

$$(18,-6)_{\mathbb{Q}} \simeq (2,-6)_{\mathbb{Q}}.$$

At $p=3$, write

$$18 = 3^2\cdot 2, \qquad -6 = 3\cdot(-2).$$

The odd-prime Hilbert-symbol formula gives

$$(18,-6)_3 = (-1)^{2\cdot 1\cdot(3-1)/2}\left(\frac{2}{3}\right)\left(\frac{-2}{3}\right)^2 = -1. \tag{13.13}$$

Thus $B\otimes\mathbb{Q}_3$ is division, and B is a division algebra over $\mathbb{Q}$.

At $p=2$, write

$$18 = 2\cdot 9, \qquad -6 = 2\cdot(-3).$$

The 2-adic formula

$$(2^\alpha u, 2^\beta w)_2 = (-1)^{((u-1)/2)((w-1)/2)+\alpha(w^2-1)/8+\beta(u^2-1)/8}$$

gives

$$(18, -6)_2 = -1. \tag{13.14}$$

At every odd prime other than 3, both valuations are zero and the Hilbert symbol is $+1$. At the real place, $18 > 0$, so the real Hilbert symbol is $+1$. Consequently

$$\mathrm{Ram}(B) = \{2, 3\}, \qquad \mathrm{disc}(B) = 6, \tag{13.15}$$

and

$$B \otimes_{\mathbb{Q}} \mathbb{R} \simeq M_2(\mathbb{R}).$$

An explicit real splitting is

$$\rho(I) = \begin{pmatrix} 3\sqrt{2} & 0 \\ 0 & -3\sqrt{2} \end{pmatrix}, \qquad \rho(J) = \begin{pmatrix} 0 & 6 \\ -1 & 0 \end{pmatrix}. \tag{13.16}$$

These matrices square to 18 and -6 and anticommute. Therefore

$$\rho(K) = \begin{pmatrix} 0 & 18\sqrt{2} \\ 3\sqrt{2} & 0 \end{pmatrix}.$$

For the element associated with (a, q, n),

$$\rho(\gamma) = \begin{pmatrix} 2 + \frac{3\sqrt{2}}{2} q & 3a + 9\sqrt{2}\, n \\ -\frac{1}{2} a + \frac{3\sqrt{2}}{2} n & 2 - \frac{3\sqrt{2}}{2} q \end{pmatrix}. \tag{13.17}$$

Its determinant is the reduced norm in (13.10).

Let $\mathcal{M}$ be a maximal overorder containing $\mathcal{O}$. Since $[\mathcal{M} : \mathcal{O}] = 18$, the finite abelian group $\mathcal{M}/\mathcal{O}$ has order 18. Lagrange's theorem therefore shows that multiplication by 18 annihilates every element of this additive quotient; equivalently, the ideal $(18) = 18\mathbb{Z}$ annihilates $\mathcal{M}/\mathcal{O}$, and $18\mathcal{M} \subset \mathcal{O}$. Hence the principal congruence subgroup

$$\mathcal{M}^1(18\mathcal{M}) := \{u \in \mathcal{M}^1 : u - 1 \in 18\mathcal{M}\}$$

satisfies the exact inclusions

$$\mathcal{M}^1(18\mathcal{M}) \subset \mathcal{O}^1 \subset \mathcal{M}^1.$$

Thus $\mathcal{O}^1$ has finite index in the norm-one group of a maximal order and contains a principal congruence subgroup. This is the required finite-index and congruence bridge; it does not replace the local type of $\mathcal{O}$ by the local type of $\mathcal{M}$.

The image

$$\bar{\Gamma} = \rho(\mathcal{O}^1)/\{\pm 1\} \subset PSL_2(\mathbb{R})$$

is a discrete congruence arithmetic Fuchsian group. Since the defining rational quaternion algebra is division, $\bar{\Gamma}\backslash\mathbb{H}$ is compact; see (Voight, 2021, Lemma 38.1.2, Proposition 38.1.3, and Definition 38.7.2).

13.5 Centralizer, optimal quadratic order, and primitivity

Fix a shell point and set

$$v = \gamma - 2.$$

Equation (13.12) gives $v^2 = 3$, hence

$$\mathbb{Q}(\gamma) = \mathbb{Q}(v) \cong \mathbb{Q}(\sqrt{3}). \tag{13.18}$$

Since γ is noncentral, the centralizer theorem for quaternion algebras gives

$$C_B(\gamma) = \mathbb{Q}(\gamma). \tag{13.19}$$

Moreover,

$$\mathbb{Z}[v] \subseteq \mathcal{O} \cap \mathbb{Q}(v).$$

Every element of $\mathcal{O}$ is integral over $\mathbb{Z}$, so every element of the intersection is an algebraic integer in $\mathbb{Q}(\sqrt{3})$. Because

$$\mathcal{O}_{\mathbb{Q}(\sqrt{3})} = \mathbb{Z}[\sqrt{3}]$$

and $\mathbb{Z}[v]$ is already isomorphic to this maximal quadratic order, both inclusions force

$$\boxed{\mathcal{O} \cap \mathbb{Q}(\gamma) = \mathbb{Z}[v] = \mathbb{Z}[\gamma - 2] \cong \mathbb{Z}[\sqrt{3}].} \tag{13.20}$$

Thus each shell point gives an optimal embedding

$$\iota_\gamma : \mathbb{Z}[\sqrt{3}] \hookrightarrow \mathcal{O}, \qquad \sqrt{3} \longmapsto \gamma - 2. \tag{13.21}$$

Conversely, every embedding (13.21) gives a trace-4, norm-1 element $2 + \iota(\sqrt{3})$, hence one integral ternary point. The shell is exactly the set of these optimal embeddings before quotienting by conjugation.

The norm-one units of $\mathbb{Z}[\sqrt{3}]$ are

$$\{\pm(2 + \sqrt{3})^k : k \in \mathbb{Z}\}. \tag{13.22}$$

For completeness, $2 + \sqrt{3}$ is the smallest positive norm-one unit greater than 1. If $x + y\sqrt{3} > 1$ has norm 1 and $y > 0$, then $y \geq 1$, $x^2 = 1 + 3y^2$, and $x + y\sqrt{3} \geq 2 + \sqrt{3}$. Multiplication by a suitable power of $2 - \sqrt{3}$ reduces any positive norm-one unit to $[1, 2 + \sqrt{3})$, proving (13.22).

Equations (13.19)–(13.22) imply

$$C_\Gamma(\gamma) = \{\pm\gamma^k : k \in \mathbb{Z}\}. \tag{13.23}$$

Hence γ is primitive in Γ and its image is primitive in $\bar{\Gamma}$. Indeed, a root of γ, or a root of its class modulo the center, would commute with γ and therefore be a unit from (13.22); the exponent equation $rk = 1$ has no solution with $k \geq 2$.

The reduced characteristic polynomial is

$$T^2 - 4T + 1,$$

so the real eigenvalues are

$$2 + \sqrt{3}, \qquad 2 - \sqrt{3}. \tag{13.24}$$

For a hyperbolic element of $PSL_2(\mathbb{R})$,

$$2\cosh(\ell/2) = |\operatorname{tr}(\rho(\gamma))| = 4.$$

Thus every shell element has primitive translation length

$$\boxed{\ell_0 = 2\operatorname{arcosh}(2) = 2\log(2 + \sqrt{3}).} \tag{13.25}$$

13.6 Integral adjoint action and finitely many classes

Let

$$\mathscr{S} = \{\gamma \in \Gamma : \operatorname{trd}(\gamma) = 4\}.$$

By (13.11), $\mathscr{S}$ is exactly the entire integral ternary shell. By (13.25), every element of $\mathscr{S}$ is hyperbolic with translation length ℓ_0.

Put

$$L = \{(a, q, n) \in \mathbb{Z}^3 : a \equiv q \pmod 2\}.$$

Conjugation preserves $\mathcal{O}$, $\mathcal{O}^0$, and reduced norm. Because

$$\operatorname{nrd}(v(a, q, n)) = \frac{3}{2}F(a, q, n),$$

it gives an exact orthogonal representation on the actual parity lattice

$$\mathrm{Ad} : \Gamma/\{\pm 1\} \hookrightarrow SO^+(L, F). \tag{13.26}$$

The kernel before dividing by the center is $\Gamma \cap \mathbb{Q}^\times = \{\pm 1\}$. No action on the larger lattice $\mathbb{Z}^3$, and no finite-index equality with its full integral orthogonal group, is asserted.

There are only finitely many Γ-conjugacy classes in $\mathscr{S}$. Choose a compact fundamental set $D \subset \mathbb{H}$ for $\bar{\Gamma}$, and fix $z_0 \in \mathbb{H}$. For each conjugacy class, conjugate a representative so that its invariant axis meets D. If $z \in D$ lies on that axis, then

$$d(z, \gamma z) = \ell_0.$$

Writing

$$R_D = \max_{w \in D} d(z_0, w),$$

the triangle inequality gives

$$d(z_0, \gamma z_0) \le 2R_D + \ell_0.$$

Proper discontinuity leaves only finitely many elements of $\bar{\Gamma}$ moving z_0 by at most this amount. Hence only finitely many $\bar{\Gamma}$-classes occur. This also proves finiteness of the corresponding Γ-classes: if two images are conjugate in $\bar{\Gamma}$, lifting a conjugator gives

$$\delta \gamma_1 \delta^{-1} = \pm \gamma_2.$$

Both sides before the possible sign have reduced trace 4; the minus sign would have reduced trace -4, so it is impossible.

Choose representatives $\gamma_1, \dots, \gamma_h$. Then

$$\boxed{\mathscr{S} = \coprod_{j=1}^{h} \{\delta \gamma_j \delta^{-1} : [\delta] \in \Gamma / C_\Gamma(\gamma_j)\}, \qquad C_\Gamma(\gamma_j) = \{\pm \gamma_j^k : k \in \mathbb{Z}\}.} \tag{13.27}$$

This proves finiteness but does not compute h or a representative list. Those finite arithmetic data are needed by a class-by-class spectral formula whose coefficients contain geodesic periods.

13.7 Exact reformulation of the Koide chamber count

For $\gamma \in \mathscr{S}$, let

$$c(\gamma) = (a(\gamma), q(\gamma), n(\gamma))$$

be the inverse coordinate map in section 13.3. Let

$$\mathcal{C} = \{(a, q, n) \in \mathbb{R}^3 : a > 0,\ q \ge 0,\ n \ge 0,\ a + 4n > q\},$$

and retain the canonical height

$$H(a, q, n) = 4a + 17n.$$

The integral bijections in theorems 2.4 and 6.3 give

$$\boxed{R(X) = \sum_{j=1}^{h} \# \left\{ [\delta] \in \Gamma / C_\Gamma(\gamma_j) : \begin{array}{l} c(\delta \gamma_j \delta^{-1}) \in \mathcal{C}, \\ H(c(\delta \gamma_j \delta^{-1})) \le X \end{array} \right\}.} \tag{13.28}$$

There is no hidden finite-to-one factor in (13.28): cosets by the centralizer parametrize the elements of each class exactly once, and (13.11) parametrizes shell elements by triples exactly once. Put $C_j = C_\Gamma(\gamma_j)$. Here Γ/C_j is the set of left cosets δC_j, not the set of right cosets $C_j\delta$, and the exact orbit bijection is

$$\Phi_j : \Gamma/C_j \longrightarrow \{\delta\gamma_j\delta^{-1} : \delta \in \Gamma\}, \qquad \delta C_j \longmapsto \delta\gamma_j\delta^{-1}.$$

Indeed, right multiplication of δ by an element of C_j does not change its conjugate, so Φ_j is well defined and surjective; and

$$\Phi_j(\delta_1 C_j) = \Phi_j(\delta_2 C_j) \iff \delta_2^{-1}\delta_1 \in C_j \iff \delta_1 C_j = \delta_2 C_j,$$

so it is injective. If $\pi : \Gamma \to \bar{\Gamma}$ is the quotient map, then

$$\Gamma/C_j \cong \bar{\Gamma}/C_{\bar{\Gamma}}(\bar{\gamma}_j)$$

via $\delta C_j \mapsto \pi(\delta)C_{\bar{\Gamma}}(\bar{\gamma}_j)$. To prove the required equality of stabilizers, observe first that $\pi(C_j) \subseteq C_{\bar{\Gamma}}(\bar{\gamma}_j)$. Conversely, if $\pi(\delta)$ centralizes $\bar{\gamma}_j$, then $\delta\gamma_j\delta^{-1} = \pm\gamma_j$. Reduced-trace invariance and $\operatorname{trd}(\gamma_j) = 4$ exclude the minus sign, so $\delta \in C_j$. Thus $\pi(C_j) = C_{\bar{\Gamma}}(\bar{\gamma}_j)$; together with $\ker\pi = \{\pm 1\} \subset C_j$, this proves the displayed bijection and shows that passage to PSL_2 introduces no factor 2. Positive trace selects the lift sign, but the chamber is not invariant under $\gamma \mapsto \gamma^{-1}$; inverse classes and their orientations must still be retained separately.

Under the splitting (13.16),

$$\|\rho(\gamma)\|_{\mathrm{F}}^2 = 8 + 9q^2 + \frac{37}{4}a^2 + \frac{105\sqrt{2}}{2}an + \frac{333}{2}n^2. \tag{13.29}$$

For $i \in \mathbb{H}$,

$$2\cosh d(i, \rho(\gamma)i) = \|\rho(\gamma)\|_{\mathrm{F}}^2.$$

Since $\sinh(\ell_0/2) = \sqrt{3}$, the standard Huber parameter is

$$X_i(\gamma)^2 = \frac{\|\rho(\gamma)\|_{\mathrm{F}}^2 - 2}{12}. \tag{13.30}$$

The shell equation simplifies (13.29) exactly:

$$\|\rho(\gamma)\|_{\mathrm{F}}^2 = 14 + \frac{(7a + 15\sqrt{2}\,n)^2}{4},$$

and hence

$$\boxed{X_i(\gamma)^2 = 1 + \frac{(7a + 15\sqrt{2}\,n)^2}{48}.} \tag{13.31}$$

Thus, in the positive chamber, the radial cutoff has the exact irrational linear height $7a+15\sqrt{2}\,n$. The Koide height $H = 4a + 17n$ is a different linear height. The two sizes are comparable:

$$a \le \frac{H}{4}, \qquad n \le \frac{H}{17}, \qquad 0 \le q < a + 4n \le \frac{33}{68}H,$$

and conversely

$$H \le \sqrt{305}\sqrt{a^2 + q^2 + n^2}.$$

Norm equivalence in the trace-zero space then compares H coarsely with X_i. Such comparability identifies the growth scale but does not transfer a main coefficient or a remainder between the sharply bounded regions.

13.8 Consequence for the open square-root discrepancy

The quaternion construction proves:

1. the ternary shell is exactly a finite union of primitive hyperbolic conjugacy classes in one explicit cocompact arithmetic Fuchsian group;
2. $\log(2 + \sqrt{3}) = \ell_0/2$, so its occurrence in the main constant is tied to the primitive translation length;
3. the conjectural exponent $1/2$ is the same exponent proposed for the standard hyperbolic lattice-point problem in conjugacy classes, giving a precise analytic analogy but not an equivalence of the two cutoff problems.

The third item classifies the remaining analytic problem; it does not prove the error estimate. Chatzakos–Petridis (Chatzakos and Petridis, 2016, Theorem 1.1) give

$$E(\mathcal{H}, X; z) = O(X^{2/3})$$

for the standard radial count after the main expression, including every small-eigenvalue term, is subtracted. Their Conjecture 5.7 proposes

$$E(\mathcal{H}, X; z) = O_\varepsilon(X^{1/2+\varepsilon}).$$

The cutoff in (13.28) uses both a sharp chamber and $4a+17n$, not the radial height $7a+15\sqrt{2}\,n$ in (13.31). Their symmetric difference can have main-order size, so even the conjectured radial bound would not by itself prove the Koide bound. Therefore the status of

$$R(X) - C_0 X = O_\varepsilon(X^{1/2+\varepsilon})$$

remains **OPEN**. Closing it requires a spectral or automorphic argument for the exact finite sum (13.28), with its exact sector and height.

13.9 Missing mathematics before an error theorem can be claimed

The bridge does not supply the following results.

1. **Class data.** Determine h, representatives γ_j, and orientation/inversion conventions. Finiteness does not compute the periods in a spectral expansion.

2. **Exact nonmaximal order.** The order has reduced discriminant 108, is nonmaximal at the ramified primes, and is not an Eichler order in the standard sense. A result for maximal or specified Eichler orders must be checked at this order or transferred through finite-index groups with all coset multiplicities retained. Index and discriminant alone do not determine the precise local order type required by a trace formula.

3. **Central quotient.** Track the passage from $\Gamma \subset SL_2(\mathbb{R})$ to $\bar{\Gamma} \subset PSL_2(\mathbb{R})$, including ± 1, inverse classes, and possible orbifold torsion.

4. **Exact cutoff.** Prove a trace formula or effective orbital theorem for all four inequalities defining $\mathcal{C}$ and for $4a + 17n \leq X$. The radial Huber cutoff is different.

5. **Sharp-boundary control.** Remove smoothing with an explicit estimate at the required scale. The two Pell faces contribute only $O(\log X)$, as already proved canonically, but this alone does not control every smoothing strip adjacent to all chamber walls.

6. **Exceptional spectrum.** Once an appropriate sector spectral expansion is constructed, one must determine whether it contains X^{s_j} terms for $\lambda_j = s_j(1 - s_j) < 1/4$, weighted by closed-geodesic periods. To subtract only $C_0 X$ with a $O(X^{1/2+\varepsilon})$ remainder, every relevant exceptional term must be shown to vanish or cancel, or it must be displayed and subtracted. No such vanishing is proved here.

7. **Pointwise square-root estimate.** After secondary terms and the exact sector are handled, the needed pointwise $1/2 + \varepsilon$ bound remains at the conjectural strength of the classical problem. Mean-square results do not give it for every X.

8. **Main coefficient.** A spectral derivation must reproduce

$$C_0 = \frac{9}{2\pi^2} \log(2 + \sqrt{3}) \log\left(\frac{3}{2\sqrt{2}}\right)$$

   using the same lattice, chamber, strict walls, height, and Haar measure. The existing local-density proof fixes this constant but does not calibrate a new trace formula automatically.

9. **Transfer between the two exact heights.** Formula (13.31) already gives the exact standard radial parameter. What is missing is a remainder-level transfer between its height $7a + 15\sqrt{2}\, n$ and the canonical height $4a + 17n$, together with the chamber walls. Coarse norm comparability does not preserve a discrepancy estimate.

These are concrete mathematical tasks. None invalidates the quaternion bridge; together they explain why it does not close the open claim.

13.10 Reproducibility and references

The canonical quaternion replay is `supporting_materials/workbench/evidence/koide_exact_checkpoint_20260904/quaternion/verify_quaternion_bridge.py` (SHA-256 `6675ba2fc4fa8c2b10e14558e412178b6b5b579c9c96afca78e1db9bcdf5c6b4`). Its recorded successful output is `supporting_materials/workbench/evidence/koide_exact_checkpoint_20260904/quaternion/verification.log` (SHA-256 `66853c6191a000c7c827aacd1e0e33dcf34dbe3dd61d746f66fa4d805f34c91e`), and its local manifest is `supporting_materials/workbench/evidence/koide_exact_checkpoint_20260904/quaternion/SHA256SUMS.txt`

(SHA-256 `f9d044db0d4b003e7aef7649b3394e6b299c09a6b2eb7adca095f8f9100ba623`). The checkpoint-wide manifest is `supporting_materials/workbench/evidence/koide_exact_checkpoint_20260904/SHA256SUMS` (SHA-256 `8d108eb928e5d07093badde169f443a4a38b6fcf3eb59d5192bdd33ecef8d7a4`). The replay checks the basis multiplication table; the trace, norm, and coordinate formulas for $(r,s,t,u) \in \{-4,-3,\dots,4\}^4$; the shell identities for $(a,q,n) \in \{-24,-23,\dots,24\}^2 \times \{-8,-7,\dots,8\}$, containing 48 shell points; the inverse identity on those points; the radial-height identity on that box; and the Hilbert-symbol formulas at 2 and 3. These finite ranges are consistency checks, not proofs outside the stated boxes. The exact canonical sources used for the counting statements imported into this bridge are:

- `supporting_materials/workbench/research/koide_counting_repair.tex`, SHA-256 `965b0a3dca8bfe4f22b9e9208368fa04697070fd25bd06bd67f2481d17cb9d9d`;
- `supporting_materials/workbench/research/koide_exact_parametrization.tex`, SHA-256 `be83ecfdf9c2eaf09c9d01ecf5a1d0aebf0b2b926f36973723b686bafc7705fc`;
- `supporting_materials/workbench/research/koide_exact_census.tex`, SHA-256 `fbb65687aa669c597baa8d1c77dfec7821f1b664a12d9b825ac6686aa1dd8a5a`.

The bounded coordinate-bijection replay, whose scope does not include the quaternion multiplication table, Hilbert symbols, centralizer, conjugacy finiteness, or spectral conclusions, is `supporting_materials/workbench/reproduction/t1_koide_exact/verify_koide_narain_root_density.py` (SHA-256 `e4bf2982b26b07add6057b5d2d5c771a52ef52c629b5ad5f0aab24e440e41986`). Its stored height-200000 output is `supporting_materials/workbench/reproduction/t1_koide_exact/koide_verification_200000.json` (SHA-256 `b907f0ee5bdd3ad80622b669722d218b79a11a5baab0d4571e141936100706b7`); the exact invocation and scope are stated in `supporting_materials/workbench/reproduction/t1_koide_exact/README.md` (SHA-256 `5e9cdcb9e67954c37d8fde69e13549172aa03245674b358e4a1d25f8d9d3bff0`). The quaternion identities and deductions in this section are proved in the displayed algebraic calculations themselves, not inferred from that bounded replay.

References:

- John Voight, *Quaternion Algebras*, Graduate Texts in Mathematics 288, especially Section 10.1, Lemma 38.1.2, and Proposition 38.1.3; DOI.
- Dimitrios Chatzakos and Yiannis N. Petridis, *The hyperbolic lattice point problem in conjugacy classes*, arXiv:1504.01307v2, Theorem 1.1 and Conjecture 5.7; arXiv.
- The exact Koide chamber, height, main term, and Pell-face estimates are in Sections 2–12 of this workbench.

14 Primary-source theorem audit of effective counting for the fixed Koide ternary quadric sector

Date: 4 September 2026
Exact counting function:

$$R(X) = \#\left\{ (a,q,n) \in \mathbb{Z}^3 : \begin{array}{l} a^2 - 3q^2 - 18n^2 = -2, \\ a > 0,\ q \geq 0,\ n \geq 0,\ a + 4n > q, \\ 4a + 17n \leq X \end{array} \right\}.$$

The leading constant proved in theorem 2.11 is

$$C_0 = \frac{9}{2\pi^2} \log(2+\sqrt{3}) \log\left(\frac{3}{2\sqrt{2}}\right).$$

The canonical project-relative mathematical sources used by this audit are:

- `supporting_materials/workbench/research/koide_counting_repair.tex`, which proves the coordinate bijection, local densities, invariant-volume factor, Gorodnik–Oh–Shah hypotheses, and calibrated constant, SHA-256 `965b0a3dca8bfe4f22b9e9208368fa04697070fd25bd06bd67f2481d17cb9d9d`;
- `supporting_materials/workbench/research/koide_exact_parametrization.tex`, which proves the exact parametrization and the Benoist–Oh power-saving application, SHA-256 `a457709dcce0fce0774eb9326b84b00d5d8144c2087f480da704974e499bcdd3`;
- `supporting_materials/workbench/research/koide_quaternion_bridge.tex`, which proves the exact order, shell, centralizer, conjugacy-class, and radial-height statements used below, SHA-256 `959989e5c38c35d5f935605c02e92b3bff9b87a026296108e0ccbc2f6a7d8f98`;
- `supporting_materials/workbench/research/koide_exact_census.tex`, which records the separately bounded finite census, SHA-256 `18ed5968b7e67d9a0b23d8675b40c37ba46170ad29efdbb9ce10ed632c99600a`.

The primary publications themselves are identified by exact versions, dates where version-sensitive, locators, and DOI or arXiv links in section 14.11. They are not claimed to be stored in a separate canonical evidence manifest.

14.1 Determination

Among the source versions enumerated in section 14.11, no theorem proves, for this exact fixed level, full integral lattice, sharp chamber, and sharp polyhedral height, the unconditional estimate

$$R(X) - C_0 X = O_\varepsilon(X^{1/2+\varepsilon}).$$

The strongest exact-domain conclusion obtained from the general theorems enumerated there is

$$\boxed{R(X) = C_0 X + O(X^{1-\eta}) \quad \text{for some unspecified } \eta > 0.}$$

It follows by applying Benoist–Oh, Definition 13.1 and Theorem 13.2, orbit by orbit after directly verifying effective well-roundedness for the present family. Benoist–Oh do not give a numerical lower bound for η, and their proof does not imply $\eta \geq 1/2$.

The superficially stronger exponents in nearby problems do not answer the question:

- Good's sharp hyperbolic conjugacy-class theorem, as formulated and reproved by Chatzakos–Petridis, gives $O(X^{2/3})$ only for a different radial cutoff and only after every small-eigenvalue term X^{s_j} has been placed in the main expression. The same paper states $O_\varepsilon(X^{1/2+\varepsilon})$ as Conjecture 5.7.

- Liu–Sarnak obtain a smooth radial error $O_\varepsilon(X^{1/2+\theta+\varepsilon})$, hence $O_\varepsilon(X^{39/64+\varepsilon})$ from $\theta = 7/64$, and square-root strength only under the generalized Selberg/Ramanujan-at-infinity conjecture for the relevant Jacquet–Langlands transfers. Their paper's global affine-sieve setup assumes $t\det(f)$ square-free, but the analytic Proposition 3.2 itself is formulated for a variable cocompact lattice with a spectral gap. After the exact quaternionic identification it applies class by class to this lattice. What it still does not supply is a sharp cutoff, the coordinate chamber, or the nonradial height $4a + 17n$.
- Gorodnik–Nevo's published sharp polar-sector theorem loses powers through set thickening. For $\dim G = 3$, it gives $O(X^{19/20+\epsilon})$ even with tempered spectrum; the Kim–Sarnak complementary-series threshold, which does apply to the exact congruence arithmetic lattice by Jacquet–Langlands, gives the still weaker $O(X^{39/40+\epsilon})$. This remains a comparison theorem because Gorodnik–Nevo's fixed polar-sector family has not been identified with the exact polyhedron, including all integral orbits.
- Square-root-scale statements in Chatzakos–Petridis, Chatzakos, Voskou, and Lekkas–Petridis are mean-square, averaged, Omega, or conjectural statements. They are not pointwise sharp-sector upper bounds.
- Recent split-stabilizer theorems assume $-m\det(F)$ is a square and have a $T\log T$-type main term. Here $-(-2)\det(F) = 108$ is not a square, so they concern the opposite arithmetic regime.

The two independent missing inputs are therefore (i) a sharp-boundary spectral or arithmetic estimate for the actual height and chamber, and (ii) control or elimination of any complementary-spectrum secondary terms for the actual sector. The Pell faces $q = 0$ and $n = 0$, which contribute only $O(\log X)$, do not solve the moving outer-boundary problem.

14.2 Exact arithmetic and geometric classification

14.2.1 The rational group and stabilizer

Let

$$F(a,q,n) = a^2 - 3q^2 - 18n^2, \qquad \mathscr{X} = \{F = -2\}.$$

The matrix of F is $\mathrm{diag}(1,-3,-18)$, so $\det(F) = 54$ and

$$-(-2)\det(F) = 108 \notin \mathbb{Q}^{\times 2}.$$

The homogeneous form F is anisotropic over $\mathbb{Q}$. Indeed, suppose a primitive integral triple satisfies

$$a^2 - 3q^2 - 18n^2 = 0.$$

Modulo 3, one gets $3 \mid a$; write $a = 3a_1$. Dividing by 3 and reducing modulo 3 gives $3 \mid q$; write $q = 3q_1$. The resulting equation is

$$a_1^2 = 3q_1^2 + 2n^2.$$

Modulo 3, this says $a_1^2 + n^2 \equiv 0 \pmod 3$, whence $3 \mid a_1, n$. Thus $3 \mid a, q, n$, contradicting primitivity.

Put $G = \mathrm{Spin}(F)$ and $x_0 = (1, 1, 0)$. The orthogonal complement of x_0 is

$$x_0^{\perp} = \{(3s, s, t) : s, t \in \mathbb{Q}\}, \qquad F(3s, s, t) = 6(s^2 - 3t^2).$$

Consequently the connected stabilizer $H = \mathrm{Stab}_G(x_0)^{\circ}$ is the norm-one torus

$$\mathrm{Res}^1_{\mathbb{Q}(\sqrt{3})/\mathbb{Q}} \mathbb{G}_m.$$

It is split over $\mathbb{R}$, anisotropic over $\mathbb{Q}$, and has no nontrivial $\mathbb{Q}$-character. Under the quaternionic identification in section 13.5, x_0 corresponds to $v_0 = (I + J)/2$, with $v_0^2 = 3$, and

$$\mathcal{O} \cap \mathbb{Q}(v_0) = \mathbb{Z}[v_0], \qquad H(\mathbb{Q}) \cap \Gamma = \{\pm(2 + v_0)^k : k \in \mathbb{Z}\}.$$

Thus $(H(\mathbb{Q}) \cap \Gamma)/\{\pm 1\}$ is infinite cyclic. Under $H(\mathbb{R})^{\circ} \cong \mathbb{R}_{>0}$, the logarithm sends its positive part to $\log(2 + \sqrt{3})\mathbb{Z} \subset \mathbb{R}$; the quotient is compact, and in particular $(H \cap \Gamma)\backslash H(\mathbb{R})$ has finite volume. Since F is $\mathbb{Q}$-anisotropic, the arithmetic Fuchsian quotient attached to G is cocompact as well. There is no ambient hyperbolic cusp.

This is the finite-volume stabilizer regime associated with a closed geodesic. It is not the rationally split regime with divergent stabilizer in which one obtains $T \log T$ growth.

14.2.2 The exact height really is a norm-sector height

Define

$$\|(a, q, n)\|_* = max\{|4a + 17n|, |q|, |a|\}.$$

This is a genuine norm because it is the pullback of the sup norm under the invertible linear map

$$(a, q, n) \longmapsto (4a + 17n, q, a),$$

whose determinant is -17. On the target chamber,

$$0 < a < 4a + 17n, \qquad 0 \le q < a + 4n < 4a + 17n,$$

and therefore $\|(a, q, n)\|_* = 4a + 17n$. Thus the outer cutoff is not merely comparable to a norm: it is exactly the restriction of a fixed ambient norm.

The additional wall $a + 4n > q$ removes only a bounded real set. If $q \ge a + 4n$, then

$$a^2 + 2 = 3q^2 + 18n^2 \ge 3(a + 4n)^2 + 18n^2,$$

so $2a^2 + 24an + 66n^2 \le 2$. Its only integral point in the positive orthant on $\mathscr{X}$ is $(1, 1, 0)$. The unbounded faces $q = 0$ and $n = 0$ are Pell families and contribute $O(\log X)$ integral points, but their real boundary strips still have to be included in a sharp-domain argument.

14.2.3 Invariant volume with a strong enough secondary estimate

Use

$$\sqrt{3}q = r\cos\theta, \qquad 3\sqrt{2}n = r\sin\theta, \qquad 0 \le \theta \le \frac{\pi}{2}.$$

On $F = -2$, $r = \sqrt{a^2+2}$, and the gauge measure is

$$dm_\infty = \frac{1}{6\sqrt{6}}\, da\, d\theta.$$

Writing $c = 17/(3\sqrt{2})$, the outer boundary is

$$4a + c\sqrt{a^2+2}\sin\theta = X.$$

If $A_X(\theta)$ is its positive solution, then uniformly in θ,

$$A_X(\theta) = \frac{X}{4 + c\sin\theta} + O(X^{-1}).$$

Indeed $A_X(\theta) \asymp X$ uniformly and $\sqrt{A^2+2} = A + O(A^{-1})$. Once X contains the bounded wall excision, it follows that

$$m_\infty(B_X) = I_\infty X + O(1), \qquad I_\infty = \frac{1}{2\sqrt{3}} \log\!\left(\frac{3}{2\sqrt{2}}\right).$$

This $O(1)$ refinement matters: a theorem whose main term is the exact invariant volume can be converted to the displayed C_0X main term without spending a power of X.

14.2.4 The full integral locus is not assumed to be one orbit

The exact lattice must not be silently replaced by a convenient single orbit or by a commensurable maximal-order lattice. Borel–Harish-Chandra finiteness, in the form quoted by Benoist–Oh on journal p. 1892, says that $G(\mathbb{Z})$ has finitely many orbits on $\mathscr{X}(\mathbb{Z})$. Thus

$$\mathscr{X}(\mathbb{Z}) = \bigsqcup_{j=1}^{h} x_j\Gamma$$

for the integral spin group Γ attached to the chosen lattice and finitely many integral representatives x_j. An effective theorem for one orbit must be applied to every $x_j\Gamma$, and the finitely many main-measure scalars must be summed. The full-lattice scalar is calibrated from Borovoi's local/Brauer–Manin density formula in theorems 2.10 and 2.11; it is not obtained by declaring that all integral points form the orbit of x_0.

14.2.5 Exact quaternionic lattice and its congruence status

For the spectral comparisons below, the precise arithmetic lattice can be retained. Put

$$B = (18,-6)_{\mathbb{Q}} = \mathbb{Q}\langle I, J\rangle/(I^2 - 18,\ J^2 + 6,\ IJ + JI), \qquad K = IJ,$$

and

$$\mathcal{O} = \mathbb{Z}1 \oplus \mathbb{Z}\frac{I+J}{2} \oplus \mathbb{Z}\frac{I-J}{2} \oplus \mathbb{Z}\frac{K}{2}.$$

The complete integral multiplication table is (13.3), and its reduced-trace Gram determinant is -108^2 by (13.6); hence $\mathrm{discrd}(\mathcal{O}) = 108$. The Hilbert-symbol calculations (13.13) and (13.14), together with the unramified-place and real-place calculation summarized in (13.15), show that the ramification set is exactly $\{2, 3\}$. Thus B is the indefinite quaternion division algebra of discriminant 6. The map

$$(a, q, n) \longmapsto \gamma(a, q, n) = 2 + \frac{qI + aJ + nK}{2}$$

is the bijection (13.11) from $F(a, q, n) = -2$ to the reduced-trace-4, reduced-norm-one locus in $\mathcal{O}$. Indeed, the parity condition $a \equiv q \pmod 2$, which is exactly what membership in $\mathcal{O}$ requires, follows automatically from $F = -2$ modulo 2, and

$$\mathrm{nrd}(\gamma) = 4 + \frac{3}{2}F(a, q, n) = 1.$$

Thus the acting Fuchsian lattice is exactly $\Gamma = \mathcal{O}^1$, not a substituted maximal-order lattice. If $\mathcal{M} \supset \mathcal{O}$ is maximal, the reduced-discriminant index formula gives

$$[\mathcal{M} : \mathcal{O}] = \frac{108}{6} = 18, \qquad 18\mathcal{M} \subset \mathcal{O},$$

and consequently

$$\mathcal{M}^1(18\mathcal{M}) \subset \mathcal{O}^1 \subset \mathcal{M}^1.$$

Here $\mathcal{M}^1(18\mathcal{M}) = \{u \in \mathcal{M}^1 : u - 1 \in 18\mathcal{M}\}$. Therefore $\mathcal{O}^1$ is a congruence arithmetic Fuchsian group in the exact sense used by Chu–Li. The number 108 is the order's reduced discriminant, not an Eichler level: the order is nonmaximal at the ramified primes $2, 3$. This non-Eichler feature changes the finite compact-open subgroup and hence which automorphic representations have fixed vectors and the dimensions of those fixed-vector spaces; it does not change the Jacquet–Langlands/Kim–Sarnak archimedean bound.

14.3 Result matrix

Source and exact locator	What is actually proved	Status for the target
Duke–Rudnick–Sarnak (1993), Theorem 1.2, p. 145; Example 1.5, pp. 146–147	One arithmetic orbit on an affine symmetric variety: count $\sim$ invariant volume; ternary anisotropic stabilizer gives linear rather than $T \log T$ growth	Correct qualitative regime, no rate and no exact sharp-sector theorem
Eskin–McMullen (1993), Theorem 1.4, author PDF p. 3	Every well-rounded sequence has count asymptotic to volume	Qualitative only; no exponent

Source and exact locator	What is actually proved	Status for the target
Gorodnik–Oh–Shah (2009), definitions before Theorem 1.6 and Theorem 1.6, p. 5; Theorem 1.18, pp. 8–9	Any norm and an admissible generic cone: full integral count $\sim$ volume	Direct qualitative theorem for the exact norm and cone, whose hypotheses are proved in theorems 2.8 and 2.9; no error term
Gorodnik–Oh–Shah (2010), Theorem 1.13, author PDF pp. 4–5	Well-rounded polar sectors and their asymptotic	Qualitative only
Benoist–Oh (2012), Theorem 1.3, p. 1894	Standard product/Euclidean height ball: relative $O(T^{-\delta})$ for some $\delta > 0$	Theorem 1.3 alone is not the exact polyhedron
Benoist–Oh (2012), Definition 13.1, p. 1931, and Theorem 13.2, pp. 1931–1932	Any independently verified effectively well-rounded family: $\mathrm{vol}(B)(1 + O(\mathrm{vol}(B)^{-\delta_0}))$	Applies to the exact family after the proof in section 14.4.2; yields only unspecified $O(X^{1-\eta})$
Gorodnik–Nevo (2012), Definition 3.1, p. 13; Theorem 8.1 and Corollary 8.3, pp. 34–35	Sharp fixed polar sectors: $O(\mathrm{vol}^{1-\zeta+\epsilon})$, $\zeta = (2n_e(p))^{-1}(1 + 3\dim G)^{-1}$	Explicit but much weaker; target polyhedron and finite orbit union require an additional reduction
Liu–Sarnak (2010), equations (2.3)–(2.8), pp. 4–5; Proposition 3.2, pp. 10–11; equation (3.65), p. 16	Smooth radial count for a cocompact lattice containing the fixed cyclic stabilizer: $O_\epsilon(T^{1/2+\theta+\epsilon})$	Exact lattice and finite class sum are available; unconditional 39/64, conditional 1/2, but fixed-width smoothing and the radial height do not match the sharp target
Good (1983), Theorem 4, p. 116, as explicitly restated and reproved in Chatzakos–Petridis (2016), Theorem 1.1, p. 3	Sharp radial hyperbolic conjugacy-class count: $O(X^{2/3})$ after all X^{s_j} small-spectrum terms	Closest sharp spectral model, not the exact polyhedron; square root is CP Conjecture 5.7
Bourgain–Kontorovich–Sarnak (2010), Theorem 1.1, p. 2	Fixed angular Fourier modes in a sharp group norm ball: $O_{n,k}(T^{4/3})$ after T^{2s_j} terms	With affine height $X \asymp T^2$, again $X^{2/3}$; constants are not uniform enough to recover a sharp interval for free
Browning–Gorodnik (2017), Theorem 2.1, p. 11; Corollaries 2.5–2.6, pp. 24–25	Full Euclidean balls, finite adelic/orbit sums, and congruence conditions: unspecified $O(\mathrm{vol}^{1-\varrho})$	Correct nonsquare homogeneous regime, wrong height geometry and no explicit exponent
Cao–Huang (2022), Theorem 1.5, p. 4; Theorem 3.4, pp. 21–22	Arithmetic purity for anisotropic nonsquare ternary quadrics; effective input for Euclidean balls has an unspecified power	Qualitative full-lattice theorem, not a square-root sector result
Cao–Huang–Zhang (arXiv:2509.23320), Setting 2.1 and Theorem 2.1	Product Euclidean/max height balls with congruence-uniform $O(\mathbf{N}(\mathfrak{l})^{\kappa}\,\mathrm{vol}^{1-\delta})$	Current extension, but δ is unspecified and the growing real factor is the standard height ball, not this cone

Source and exact locator	What is actually proved	Status for the target
Oh–Shah (2014), Theorem 1.2, p. 3; Theorem 6.1, pp. 13–14	Divergent-geodesic/split-stabilizer $T \log T$ count; smooth error $O(T)$, sharp relative logarithmic error	Excluded by the finite-volume anisotropic stabilizer
Xu–Zhang (2023), main theorem; Huang (arXiv:2412.03350), Theorem 1.1; Zhang (arXiv:2603.21080), Theorem 1.1	Split-stabilizer results assuming $-m\det(F)$ is a square, with smooth or radial $T \log T$-type expansions	108 is nonsquare; wrong main-term regime
Huang (arXiv:2412.05903), Theorems 1.1–1.2	Smooth compact real weight while the level varies as $m_0 p_0^{2h}$; error of logarithmic relative strength	Varying level and smooth weight, not the fixed-level sharp count
Li–Liu (arXiv:2501.15033v2), Theorems 1.1–1.2, pp. 2–3; Lemma 2.3, pp. 5–6	Almost-prime results using a smooth radial orbit count with $O_\epsilon(d^{1+\epsilon}T^{1/2+\theta+\epsilon})$	Current use of the Liu–Sarnak input; globally assumes $t\det(f)$ square-free and supplies no sharp-sector count
Voskou (arXiv:2407.03134v3), Theorem 3 and Conjecture 1, p. 3	Four sharp orientation sectors in a closed-geodesic double-coset problem: $O(X^{2/3})$; square root conjectured	Current sharp comparator, not the target
Lekkas–Petridis (arXiv:2509.12902, version dated 24 Aug. 2026), Theorem 1.2, Theorem 1.5, Conjecture 1.6	Sharp closed-geodesic double-coset $O(X^{2/3})$, square-root mean square, pointwise square-root conjecture	Version-specific confirmation that mean square does not furnish the requested pointwise bound

14.4 Benoist–Oh: exact applicability and exact limitation

Among the general effective theorems enumerated in section 14.3, this is the one whose stated hypotheses can be verified for the actual family without changing its height or chamber.

14.4.1 What the theorem assumes

Benoist–Oh work with a connected semisimple simply connected almost K-simple group G, a finite-kernel rational representation, a symmetric subgroup H, and the hypothesis that H° has no nontrivial K-character; see journal pp. 1892–1893. For a congruence subgroup Γ_S, Theorem 12.5, journal p. 1929, supplies the effective equidistribution of the translates of $(\Gamma_S \cap H_S)\backslash H_S$ used in Theorem 13.2. Example (A), journal p. 1896, explicitly treats ternary quadratic forms provided the homogeneous form does not represent zero over $\mathbb{Q}$.

All of these hypotheses hold here:

- $G = \mathrm{Spin}(F)$ is simply connected and absolutely almost simple of type A_1, and $G(\mathbb{R})$ is noncompact.

- The standard three-dimensional orthogonal representation has finite kernel.
- The stabilizer of a nonisotropic vector is symmetric.
- The mod-3 descent in section 14.2.1 proves that F does not represent zero over $\mathbb{Q}$.
- H° is the $\mathbb{Q}$-anisotropic norm-one torus of $\mathbb{Q}(\sqrt{3})$, so it has no nontrivial $\mathbb{Q}$-character and its arithmetic quotient is compact.
- With $S = \{\infty\}$, the full integral spin group is itself a congruence subgroup in Benoist–Oh's sense.

There is no omitted wavefront or properness condition: symmetry of H is exactly what invokes their effective wavefront argument in Lemma 12.3 and Theorem 12.5. Cocompactness makes escape in the automorphic quotient simpler, but does not remove Definition 13.1(3), which still has to be checked on the growing affine set.

Here "congruence" is an exact property of the chosen integral group, not an inference from mere commensurability. Under the quaternionic realization of G, the finite-integrality conditions define a compact-open subgroup of $G(\mathbb{A}_f)$; equivalently, after placing the corresponding order $\mathcal{O}$ in a maximal order $\mathcal{O}_{\max}$, some principal subgroup $\{u \in \mathcal{O}^1_{\max} : u \equiv 1 \pmod{N\mathcal{O}_{\max}}\}$ lies in the exact integral spin group. This distinction matters for spectral-gap transfer: an arbitrary finite-index noncongruence subgroup need not inherit the Kim–Sarnak gap.

14.4.2 Definition 13.1 for the actual B_X

Let $B_X \subset \mathscr{X}(\mathbb{R})$ be the exact chamber cut out by the five linear conditions and $4a + 17n \le X$. In Benoist–Oh's notation, $B^+_{X,\epsilon} = B_X B_\epsilon$ and $B^-_{X,\epsilon} = \bigcap_{u \in B_\epsilon} B_X u$.

Condition (1) is void because there is no finite-place factor.

For condition (2), if u lies in an ϵ-ball about the identity, every fixed linear functional L defining the chamber or height satisfies

$$|L(xu) - L(x)| \ll \epsilon \|x\|_2.$$

On the unbounded portion of this chamber, $\|x\|_2 \asymp 1 + a \asymp 1 + (4a + 17n)$. Hence a membership change under u is confined to:

1. an angular strip of width $O(\epsilon)$ at $q = 0$ or $n = 0$;
2. an $O(\epsilon X)$-thick radial shell at $4a + 17n = X$; or
3. an $O(\epsilon)$-neighborhood of one of the compact boundary pieces $a = 0$ or $a + 4n = q$.

Since $dm_\infty = (6\sqrt{6})^{-1} da\, d\theta$, their total measure is $O(\epsilon X)$. Thus

$$m_\infty(B^+_{X,\epsilon} \setminus B^-_{X,\epsilon}) \ll \epsilon\, m_\infty(B_X),$$

so Definition 13.1(2) holds with $\kappa = 1$.

For condition (3), Benoist–Oh's Euclidean height is comparable to $1 + a$ throughout a fixed group-neighborhood enlargement of the chamber. For every $k > 0$, uniformly in $0 < \epsilon < 1$,

$$\int_{B^+_{X,\epsilon}} H_\infty(x)^{-k}\, dm_\infty(x) \ll_k \int_0^{O(X)} (1+a)^{-k}\, da.$$

The right side is $O(X^{1-k})$ for $0 < k < 1$, $O(\log X)$ for $k = 1$, and $O(1)$ for $k > 1$. As $m_\infty(B_X) \asymp X$, this is $O(m_\infty(B_X)^{1-\delta(k)})$ for some $\delta(k) > 0$ in every case. Definition 13.1(3) is therefore satisfied.

Proposition 14.2 of Benoist–Oh verifies only their standard product height balls. It should not be cited as though it automatically covers this polyhedron. The preceding direct verification is the necessary bridge to Theorem 13.2.

14.4.3 Finite-orbit passage and main-term normalization

Benoist–Oh state on journal p. 1892 that $G(\mathcal{O}_S)$ has finitely many orbits on $\mathscr{X}(\mathcal{O}_S)$. Apply Theorem 13.2 to each exact integral orbit. If $\eta_j > 0$ is the exponent for the j-th orbit, take $\eta = \min_j \eta_j > 0$. The finite sum has the form

$$R(X) = \kappa_{\mathrm{orb}}\, m_\infty(B_X) + O(X^{1-\eta}).$$

Here κ_{orb} is the sum of the orbit-volume normalization constants. The comparison with Borovoi's Theorem 1.4.1 and Proposition 1.5.1 is carried out in theorem 2.11; specifically, (2.28), (2.40), and (2.62) give

$$\kappa_{\mathrm{orb}} = \mathfrak{S}(\mathscr{X}) = \frac{9\sqrt{3}}{\pi^2}\log(2+\sqrt{3}), \qquad \kappa_{\mathrm{orb}} I_\infty = C_0.$$

Together with $m_\infty(B_X) = I_\infty X + O(1)$, this yields

$$R(X) = C_0 X + O(X^{1-\eta}).$$

This counts the actual integral lattice. No assertion that $\mathscr{X}(\mathbb{Z})$ is one orbit, and no replacement by a maximal order, is involved.

14.4.4 Why the exponent remains unspecified

The proof of Benoist–Oh Theorem 13.2 balances an effective-equidistribution/Sobolev error and a boundary-thickening error:

$$c_1 \epsilon^{-p} v_X^{1-\delta} + c_2 \epsilon^{\kappa} v_X, \qquad v_X = \mathrm{vol}(B_X).$$

The resulting saving is of the form

$$\eta = \frac{\delta\kappa}{p+\kappa} > 0.$$

Neither the Sobolev exponent p nor the effective equidistribution exponent δ is made numerically strong enough in the theorem to show $\eta \geq 1/2$. Thus Benoist–Oh supplies a genuine exact-family power saving, but not the requested square-root bound.

14.5 The sharp hyperbolic frontier

14.5.1 Good and Chatzakos–Petridis

For a cocompact or cofinite Fuchsian group and a fixed hyperbolic conjugacy class $\mathcal{H}$, Chatzakos–Petridis define, in equations (1.4)–(1.6),

$$M(\mathcal{H}, X; z) = \sum_{1/2<s_j\leq 1} A(s_j)\widehat{u}_j u_j(z) X^{s_j},$$

where $\lambda_j = s_j(1-s_j) < 1/4$ and $\widehat{u}_j$ is the period of u_j along the invariant closed geodesic. Their Theorem 1.1, author manuscript p. 3, restates Good's Theorem 4, p. 116, and proves

$$N(\mathcal{H}, X; z) - M(\mathcal{H}, X; z) = O(X^{2/3}).$$

The $2/3$ arises from the sharp majorant/minorant error

$$O(XY^{-1/2} + Y + X^{1/2}\log X)$$

after choosing $Y = X^{2/3}$. Theorems 1.2–1.3 are mean-square results, not pointwise improvements. Conjecture 5.7, author manuscript p. 22, explicitly asks for

$$O_\varepsilon(X^{1/2+\varepsilon}).$$

There are two independent failures of direct transfer to $R(X)$:

1. the Huber cutoff is radial, whereas $4a + 17n \leq X$ plus the chamber is a nonradial sharp polyhedron; norm comparability does not preserve the leading constant or error because the symmetric difference can have volume of order X;
2. even in the radial problem, the theorem subtracts every nonzero small-eigenvalue contribution, not only the linear term.

For the exact Koide sector no spectral expansion establishing actual X^{s_j} terms has yet been produced. They are therefore a necessary obstruction to check, not terms that this audit claims are already known to occur. A proof of $R(X) - C_0 X = O(X^{1/2+\varepsilon})$ would have to show absence or vanishing of all relevant complementary-spectrum coefficients, or explicitly subtract them and then prove they vanish in the target formulation.

14.5.2 Current sharp refinements remain at $2/3$

Voskou, Theorem 3 of arXiv:2407.03134v3, proves $O(X^{2/3})$ for four prescribed orientation/side sectors in a closed-geodesic-to-closed-geodesic problem, with all small-spectrum period terms included in the main expression. Conjecture 1 on the same page proposes $O_\varepsilon(X^{1/2+\varepsilon})$, while Theorem 4 is a mean-square result.

Lekkas–Petridis, in the 24 August 2026 version of arXiv:2509.12902, Theorem 1.2, likewise prove

$$N(X,l)=\sum_{1/2<s_j\le 1}\frac{2}{\pi}\gamma_1(s_j)|\widehat{u}_j|^2X^{s_j}+O(X^{2/3})$$

for a closed-geodesic double-coset count. Their Theorem 1.5 gives

$$\frac{1}{X}\int_X^{2X}|E(x,l)|^2\,dx\ll X\log^2 X,$$

and Conjecture 1.6 proposes the pointwise $X^{1/2+\epsilon}$ bound. Their introduction states that Good's 2/3 has not been improved for any of the nine point/geodesic/cusp double-coset configurations. This is strong current evidence about the sharp frontier, but neither paper treats the exact Koide height.

14.5.3 Fixed angular harmonics do not give a sharp interval for free

Bourgain–Kontorovich–Sarnak, Theorem 1.1, prove for fixed left and right K-Fourier modes n,k

$$\sum_{\|\gamma\|<T}e^{2in\theta_1(\gamma)}e^{2ik\theta_2(\gamma)}=\mathbf{1}_{n=0}\mathbf{1}_{k=0}c_0T^2+\sum_j c_j(n,k)T^{2s_j}+O_{n,k}(T^{4/3}).$$

On the adjoint orbit, affine height is quadratic in the group norm, so $T^{4/3}$ again corresponds to $X^{2/3}$. Recovering a sharp angular interval requires summing many Fourier modes, while the theorem's implied constants depend on n,k. It is not a uniform desmoothing theorem for this chamber.

14.6 Smooth spectral counting

Liu–Sarnak's paper is the closest source with a square-root-plus-gap exponent. Its global arithmetic setup and the hypotheses of its analytic proposition must be separated.

The introduction, author PDF p. 2, fixes an indefinite anisotropic integral ternary form f, a nonzero level t, and assumes $t\det(f)$ is square-free. Thus the paper's affine-sieve theorem, as globally stated, does not cover $t\det(f)=-108$. Equations (2.3)–(2.4), author PDF p. 4, fix one arithmetic orbit $\mathcal{O}$ and a smooth K-invariant radial weight F_T satisfying

$$F_T=1\quad(|x|\le T/c_0),\qquad F_T=0\quad(|x|\ge c_0T),$$

with $c_0>1$ fixed. In Fermi coordinates, Proposition 3.2 uses a smooth even function equal to one for $|\rho|\le R-1$, zero for $|\rho|\ge R+1$, and with uniformly bounded derivatives. Unlike the global sieve statement, Proposition 3.2 is formulated for any variable subgroup $\Gamma<\mathrm{SL}_2(\mathbb{R})$ satisfying its uniform local condition (3.1), the spectral gap (2.7), and containing the fixed primitive cyclic group Γ_ξ. It proves

$$H_R(z)=\frac{X(R)}{\mathrm{Area}(\Gamma\backslash\mathbb{H})}+O_\varepsilon(e^{(1/2+\theta+\varepsilon)R}),\qquad X(R)\asymp e^R.$$

Equation (3.60) has $T=\sinh R$, and equation (3.65) transfers this to the one-sheeted affine orbit. Thus the smooth affine error is

$$O_\varepsilon(T^{1/2+\theta+\varepsilon}).$$

Equation (2.7) assumes $\lambda_1 \geq 1/4 - \theta^2$. The paper invokes Jacquet–Langlands and Kim–Sarnak to take $\theta = 7/64$, giving

$$O_\varepsilon(T^{39/64+\varepsilon})$$

unconditionally in its congruence-arithmetic setting; $\theta = 0$, and hence square root, is conditional on the generalized Selberg/Ramanujan-at-infinity conjecture for the relevant Jacquet–Langlands transfers.

The numerical input itself is Kim–Sarnak, Appendix 2, Proposition 2, printed p. 176. It applies to an arbitrary cuspidal automorphic representation of $\mathrm{GL}_2/\mathbb{Q}$: for a spherical real component, $|\Re\mu_{j,\infty}| \leq 7/64$, independently of finite conductor. Jacquet–Langlands preserves the real component because the quaternion algebra is split at infinity. Chu–Li, proof of Corollary 1.2, equation (5.2), printed p. 5126, records the resulting

$$\lambda_1 \geq \frac{975}{4096}$$

for every congruence arithmetic Fuchsian group. Thus the nonmaximal or non-Eichler structure of the exact finite order does not invalidate $s \leq 39/64$. It changes the finite compact-open subgroup and hence which automorphic representations have fixed vectors and the dimensions of those fixed-vector spaces; it does not change this archimedean bound. This validates the spectral-gap input for the exact lattice. It does not provide sharp-boundary control or force the periods of possible complementary-spectrum eigenfunctions to vanish.

For the exact order lattice $\Gamma = \mathcal{O}^1$, each primitive trace-four conjugacy class supplies the cyclic subgroup in Proposition 3.2, and there are only finitely many such classes. Hence the proposition and equation (3.65) do give an unconditional $39/64$ estimate after summing the corresponding **smooth standard-radial** counts. The square-free hypothesis is not an obstruction to this isolated analytic application.

This still does not desmooth to the target. A transition of fixed width in R is a fixed multiplicative annulus in T, whose invariant volume is of order T, not $T^{1/2}$. Shrinking the transition makes the derivative/Sobolev constants in Proposition 3.2 grow, and the paper gives no uniform estimate that balances this loss at square-root scale. It also has no sharp coordinate chamber, and its radial height is not $4a + 17n$. These geometric failures remain even after the exact lattice and finite class sum have been supplied.

The same distinction applies to any informal statement that a tempered spectral component has square-root size. Duke–Rudnick–Sarnak's discussion on p. 155 estimates individual tempered frequencies; it does not control the complete spectral sum evaluated at a point with a shrinking approximation to a sharp characteristic function.

14.7 General effective sector and full-lattice theorems

14.7.1 Gorodnik–Nevo's explicit generic exponent

Gorodnik–Nevo Definition 3.1 defines

$$n_e(2) = 1, \qquad n_e(p) = \text{the least even integer at least } p/2 \quad (p > 2).$$

Their Section 8 considers

$$S_t(\Phi,\Psi)=\{g:\log\|gv_0\|\le t\}\cap\Phi A^+\Psi.$$

Theorem 8.1, author PDF p. 34, assumes L^{p+} matrix coefficients and Lipschitz-well-rounded fixed angular sets; Corollary 8.3, p. 35, passes to an orbit when $(\Gamma\cap H)\backslash H$ is compact. The exponent is

$$\zeta=\frac{1}{2n_e(p)(1+3\dim G)}.$$

For $\dim G=3$,

$$\zeta=\frac{1}{20n_e(p)}.$$

Even tempered spectrum has $p=2$, $n_e(p)=1$, and hence absolute error $O(X^{19/20+\epsilon})$. For the exact congruence arithmetic lattice, Jacquet–Langlands and Kim–Sarnak give $s\le 39/64$. The corresponding matrix coefficients are L^{p+} for $p=64/25$, so $n_e(p)=2$ and the theorem gives only $O(X^{39/40+\epsilon})$.

This numerical calculation is an upper benchmark for the published generic polar-sector theorem, not a claim that Corollary 8.3 literally counts $R(X)$. A fixed polar sector contains every radial parameter in fixed polar directions. An arbitrary affine linear cone and its finite-radius polyhedral height need an exact polar decomposition or a quantitatively controlled symmetric-difference argument; all integral orbits must then be included. None of those identifications appears in the theorem statement.

Gorodnik–Nevo themselves note after Corollary 5.3, author PDF p. 23, that their sharp-set counting mechanism remains worse than square-root volume error even for tempered spectrum, whereas square-root quality is available there only for a smooth weighted problem.

14.7.2 Browning–Gorodnik and subsequent full-lattice work

Browning–Gorodnik Theorem 2.1 gives, for a Euclidean ball B_H in one congruence orbit,

$$|\Gamma_\ell y\cap B_H|=\frac{m_{Z_{y,\ell}}(Z_{y,\ell})}{m_{X_\ell}(X_\ell)}m_Y(B_H)+O\Big(\ell^{\dim H+\dim G}m_Y(B_H)^{1-\varrho}\Big)$$

for some unspecified $\varrho>0$. Corollaries 2.5–2.6 sum adelic and integral orbits and retain the Brauer/Kottwitz density, at the cost of $\ell^{\dim H+2\dim G}$. This is a useful exact-lattice precedent, but its archimedean set is the full norm ball, not the present sector, and no square-root exponent is asserted.

Cao–Huang Theorem 1.5 proves arithmetic purity of the Hardy–Littlewood property for an anisotropic ternary affine quadric with $-m\det(q)$ nonsquare. Its Theorem 3.4 invokes Nevo–Sarnak, Gorodnik–Nevo, and Browning–Gorodnik to furnish an unspecified positive exponent, denoted β_X there with X meaning the affine variety; write it here as $\beta_{\mathscr{X}}>0$ to distinguish it from the height variable. It is not a sharp-sector remainder theorem.

The current Cao–Huang–Zhang preprint, arXiv:2509.23320, Setting 2.1 and Theorem 2.1, gives a congruence-uniform power saving

$$O\Big(\mathbf{N}(\mathfrak{l})^{\kappa}\,\mathrm{vol}(\mathbb{B}_S(N))^{1-\delta}\Big)$$

for standard product Euclidean/max height balls and a compact piecewise-smooth factor. The exponent δ is unspecified, and the unbounded real factor in the target is neither compact nor their standard Euclidean ball. This preprint extends the general effective framework but does not improve the conclusion of the direct Benoist–Oh argument in sections 14.4.2 and 14.4.3.

14.8 Results in different arithmetic regimes

14.8.1 Square discriminant and divergent stabilizer

Oh–Shah Theorem 1.2 assumes

$$\mathrm{vol}((H\cap\Gamma)\backslash H)=\infty$$

and obtains a smoothed $cT\log T+O(T)$ count. Its sharp version has only a logarithmic relative error. Theorem 6.1 allows an open cone and arbitrary norm, but retains the logarithmic main-factor characteristic of its divergent-stabilizer regime; that regime is excluded here.

Xu–Zhang, *Quarterly Journal of Mathematics* 74 (2023), 659–685, assumes $-m\det(f)$ is a square over the number field. Huang, arXiv:2412.03350, Theorem 1.1, assumes the same split condition and a fixed smooth compact weight; its principal error is $B(\log B)^{1983/1984}$ in a $B\log B$ count. Zhang, arXiv:2603.21080, Section 1.1 and Theorem 1.1, again assumes $-m\det(q)\in(k^{\times})^2$ and proves a smooth K_0-invariant expansion with an error $O(T^{l_1+l_2-\delta})$, $0<\delta<1/2$; Remark 1.2 obtains an authentic count for $\Gamma=\mathrm{SL}_2(\mathcal{O}_k)$. These are important current results, but all lie in the rationally split torus regime excluded by $108\notin\mathbb{Q}^{\times 2}$, and none uses the target polyhedral height.

14.8.2 Varying level, isotropic forms, and special rotational quadrics

Huang, arXiv:2412.05903, studies $F(x)=m_0N$ with $N=p_0^{2h}\to\infty$, a smooth compactly supported real weight $w(x/\sqrt{N})$, and prescribed local conditions. Theorems 1.1–1.2 have a $\sqrt{N}/(\log N)^A$ error. This is a Linnik-type varying-level theorem, not a fixed affine shell.

Li–Liu, arXiv:2501.15033v2 (18 April 2025), Theorems 1.1–1.2, prove new almost-prime conclusions for anisotropic ternary equations. Their global assumption remains $t\det(f)$ square-free. The quantitative counting input in Lemma 2.3, pp. 5–6, is

$$O_{\varepsilon}(d^{1+\varepsilon}T^{1/2+\theta+\varepsilon})$$

for the same kind of smooth radial orbit weight inherited from Liu–Sarnak. It is a current application of the 7/64 gap, not a new unsmoothed sector theorem.

Lowry-Duda's thesis, arXiv:1704.02376, Theorem 5.5.4, treats the special rotational form $x_1^2+x_2^2=x_3^2+h$ in a full Euclidean ball. Written in ordinary radius T, its three-dimensional sharp error is $O(T^{21/22+\epsilon})$. The homogeneous form is $\mathbb{Q}$-isotropic, and neither the lattice nor the height sector is the target one.

Results on values of irrational ternary forms, shrinking sets while the discriminant varies, or positive-definite ternary representation numbers answer different problems. Lindelöf or

generalized Lindelöf hypotheses occurring there do not yield the present fixed-level pointwise sector bound.

14.9 Spectral-gap, GRH, smoothing, and averaging taxonomy

Classification	Source result	Why it does not prove the target
Unconditional, exact sharp family	Benoist–Oh Theorem 13.2 plus the direct Definition 13.1 verification	Gives only $O(X^{1-\eta})$ with $\eta > 0$ unspecified
Unconditional, exact norm/cone, full lattice	Gorodnik–Oh–Shah Theorem 1.6, with the exact hypotheses proved in theorems 2.8 and 2.9	Qualitative $C_0X + o(X)$ only
Unconditional, sharp radial closed-geodesic count	Good / Chatzakos–Petridis	$O(X^{2/3})$, different height, and main expression includes small-spectrum terms
Unconditional, smooth radial count	Liu–Sarnak Proposition 3.2 with $\theta = 7/64$, summed over the exact finite conjugacy-class decomposition	$39/64$ for the exact lattice, but fixed multiplicative smoothing and a different radial height
Selberg/Ramanujan conditional	Liu–Sarnak with $\theta = 0$	Square-root only for the smooth radial problem; still not the exact count
GRH/Lindelöf conditional	None among the source versions in section 14.11 proves this fixed sharp-sector estimate	The occurrences listed there concern varying levels/forms, shrinking-set equidistribution, or lower/Omega results
Smoothed	Liu–Sarnak; Oh–Shah; Huang I; Zhang 2026	Smoothing or split arithmetic changes the theorem; no square-root desmoothing supplied
Averaged/mean square	Chatzakos–Petridis Theorems 1.2–1.3; Voskou Theorem 4; Lekkas–Petridis Theorem 1.5	Does not imply a uniform pointwise estimate at every X
Omega/lower-bound	Chatzakos (2019), including a cocompact geodesic-average $\Omega(X^{1/2}\log\log\log X)$	Evidence for scale, not an upper bound
Isotropic or split	Oh–Shah; Xu–Zhang; Huang I; Zhang 2026; Lowry-Duda	Different stabilizer, quotient, main order, form, or level
Varying level	Huang II	Counts $F = m_0 p_0^{2h}$, not $F = -2$

In particular, "assuming GRH" is not a substitute for the missing theorem. The conditional hypothesis that occurs naturally in the closest smooth spectral result is the Selberg/Ramanujan

eigenvalue conjecture, and even it only reaches square root before the sharp-boundary unsmoothing problem is addressed.

14.10 Citation-ready conclusion

Subject to the bounded-search qualification in section 14.12, the following paragraph records the conclusion supported by the enumerated source versions:

> Among the source versions enumerated in section 14.11, none implies the unconditional sharp estimate $R(X) - C_0X = O_\varepsilon(X^{1/2+\varepsilon})$ for the fixed ternary quadric $a^2 - 3q^2 - 18n^2 = -2$ in the chamber $a > 0$, $q, n \geq 0$, $a + 4n > q$, with height $4a + 17n \leq X$. Gorodnik–Oh–Shah, Theorem 1.6, applies to the exact admissible generic cone and an arbitrary norm—with those exact hypotheses proved in theorems 2.8 and 2.9—but gives only a qualitative asymptotic. Benoist–Oh, Definition 13.1 and Theorem 13.2, applies after the direct effective-well-roundedness verification in section 14.4.2 and, after summing the finitely many integral spin-group orbits, yields $R(X) = C_0X + O(X^{1-\eta})$ for some unspecified $\eta > 0$, not $\eta = 1/2$. In the closest sharp closed-geodesic problem, Good's theorem, as restated and reproved by Chatzakos–Petridis, Theorem 1.1, gives $O(X^{2/3})$ only after all small-eigenvalue terms X^{s_j} are included; Chatzakos–Petridis, Conjecture 5.7, states the $O_\varepsilon(X^{1/2+\varepsilon})$ improvement as a conjecture. Liu–Sarnak, Proposition 3.2, applies class by class to the exact congruence quaternionic lattice and proves the smooth radial bound $O_\varepsilon(X^{1/2+\theta+\varepsilon})$, unconditionally with $\theta = 7/64$ and conditionally with $\theta = 0$, but its fixed-width smoothing and radial height do not match the exact sharp count. Thus complementary-spectrum terms and sharp nonradial boundary control remain separate unresolved inputs.

14.11 Primary references and exact locators

1. Yves Benoist and Hee Oh, "Effective equidistribution of S-integral points on symmetric varieties," *Annales de l'Institut Fourier* **62** (2012), 1889–1942. DOI, official article/PDF, arXiv. Standing setting pp. 1892–1893; finite orbit statement p. 1892; Theorem 1.3 p. 1894; Theorem 12.5 p. 1929; Definition 13.1 p. 1931; Theorem 13.2 pp. 1931–1932; Proposition 14.2 p. 1933; ternary Example (A) p. 1896.

2. Alexander Gorodnik, Hee Oh, and Nimish Shah, "Integral points on symmetric varieties and Satake compactifications," *American Journal of Mathematics* **131** (2009), 1–57. DOI, arXiv, author PDF. Admissible/generic definitions and Theorem 1.6 pp. 4–5; Theorem 1.18 pp. 8–9.

3. Alexander Gorodnik, Hee Oh, and Nimish Shah, "Strong wavefront lemma and counting lattice points in sectors," *Israel Journal of Mathematics* **176** (2010), 419–444. DOI, arXiv. Theorem 1.13, author PDF pp. 4–5.

4. Alexander Gorodnik and Amos Nevo, "Counting lattice points," *Journal für die reine und angewandte Mathematik* **663** (2012), 127–176. DOI, arXiv, author PDF. Theorem 1.9 pp. 4–5; Definition 3.1 p. 13; Theorem 4.5 pp. 17–18; Theorem 8.1 and Corollary 8.3 pp. 34–35; Proposition 8.4 p. 35.

5. William Duke, Zeév Rudnick, and Peter Sarnak, "Density of integer points on affine homogeneous varieties," *Duke Mathematical Journal* **71** (1993), 143–179. DOI, author

PDF. Theorem 1.2 p. 145; Example 1.5 pp. 146–147; discussion of individual tempered frequencies p. 155; Theorem 3.5 p. 161.

6. Alex Eskin and Curtis McMullen, "Mixing, counting, and equidistribution in Lie groups," *Duke Mathematical Journal* **71** (1993), 181–209. DOI, author PDF. Theorem 1.4, author PDF p. 3; Theorems 6.1 and 6.4 pp. 30–31; Open Question 1 p. 35.

7. Jianya Liu and Peter Sarnak, "Integral points on quadrics in three variables whose coordinates have few prime factors," *Israel Journal of Mathematics* **178** (2010), 393–426. DOI, author PDF. Arithmetic hypothesis p. 2; equations (2.3)–(2.8) pp. 4–5; Proposition 3.2 pp. 10–11; equations (3.60) and (3.65) pp. 15–16.

8. Anton Good, *Local Analysis of Selberg's Trace Formula*, Lecture Notes in Mathematics 1040, Springer, 1983. DOI. Theorem 4, p. 116. The theorem statement and a new proof were checked through Chatzakos–Petridis (2016) because the monograph text was not directly available in this audit.

9. Dimitrios Chatzakos and Yiannis N. Petridis, "The hyperbolic lattice point problem in conjugacy classes," *Forum Mathematicum* **28** (2016), 981–1003. DOI, arXiv, author manuscript. Equations (1.4)–(1.6) and Theorem 1.1 p. 3; Theorems 1.2–1.3 pp. 3–4; Conjecture 5.7 p. 22.

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11. Jean Bourgain, Alex Kontorovich, and Peter Sarnak, "Sector estimates for hyperbolic isometries," *Geometric and Functional Analysis* **20** (2010), 1175–1200. DOI, arXiv. Theorem 1.1, p. 2.

12. Hee Oh and Nimish Shah, "Limits of translates of divergent geodesics and integral points on one-sheeted hyperboloids," *Israel Journal of Mathematics* **199** (2014), 915–931. DOI, arXiv. Theorem 1.2 p. 3; Corollary 1.4 pp. 3–4; Theorem 6.1 pp. 13–14.

13. Tim D. Browning and Alexander Gorodnik, "Power-free values of polynomials on symmetric varieties," *Proceedings of the London Mathematical Society* **114** (2017), 1044–1080. DOI, arXiv. Theorem 2.1 p. 11; Corollaries 2.5–2.6 pp. 24–25.

14. Yang Cao and Zhizhong Huang, "Arithmetic purity of the Hardy–Littlewood property and geometric sieve for affine quadrics," *Advances in Mathematics* **398** (2022), 108236. DOI, arXiv. Theorem 1.5 p. 4; Proposition 2.6 pp. 12–13; Theorem 3.4 pp. 21–22.

15. Yang Cao, Zhizhong Huang, and Runlin Zhang, "Effective equidistribution, arithmetic purity of strong approximation, and geometric sieve for affine quadrics," arXiv:2509.23320. Setting 2.1 and Theorem 2.1; arXiv HTML retrieved 4 September 2026.

16. Mikhail Borovoi, "On representations of integers by indefinite ternary quadratic forms," *Journal of Number Theory* **90** (2001), 281–293. DOI, arXiv, author PDF. Theorem 1.4.1 and Proposition 1.5.1; qualitative local/Brauer–Manin coefficient, no error exponent.

17. Fei Xu and Runlin Zhang, "Counting integral points on indefinite ternary quadratic equations over number fields," *Quarterly Journal of Mathematics* **74** (2023), 659–685. DOI, arXiv. Square-discriminant hypothesis in the main theorem.

18. Zhizhong Huang, "Quantitative strong approximation for ternary quadratic forms I," arXiv:2412.03350. Theorem 1.1; square-discriminant, smooth fixed-weight $B \log B$ count.

19. Zhizhong Huang, “Quantitative strong approximation for ternary quadratic forms II,” arXiv:2412.05903. Theorems 1.1–1.2; varying level $m_0p_0^{2h}$ and smooth compact weight.

20. Runlin Zhang, “Effective count of integer points on ternary affine quadrics and effective equidistribution,” arXiv:2603.21080. Section 1.1 and Theorem 1.1; split-stabilizer smooth count; Remark 1.2 on the authentic radial count.

21. Marios Voskou, “Refined Counting of Geodesic Segments in the Hyperbolic Plane,” arXiv:2407.03134v3. Theorems 3–4 and Conjecture 1, pp. 3–4.

22. Dimitrios Lekkas and Yiannis N. Petridis, “A relative trace formula and counting geodesic arcs in the hyperbolic plane,” arXiv:2509.12902, version dated 24 August 2026. Theorem 1.2, Theorem 1.5, and Conjecture 1.6.

23. Henry H. Kim, “Functoriality for the exterior square of GL_4 and the symmetric fourth of GL_2,” with Appendix 2 by Henry H. Kim and Peter Sarnak, “Refined estimates towards the Ramanujan and Selberg conjectures,” *Journal of the American Mathematical Society* **16** (2003), 139–183. DOI, official AMS PDF. Appendix 2, Proposition 2, printed p. 176: $|\Re\mu_{j,\infty}| \le 7/64$ for a spherical real component of any cuspidal $\mathrm{GL}_2/\mathbb{Q}$ representation, and the displayed consequence $\lambda_1 \ge 975/4096$.

24. Michelle Chu and Han Li, “Small generators of cocompact arithmetic Fuchsian groups,” *Proceedings of the American Mathematical Society* **144** (2016), 5121–5127. DOI, arXiv. Section 3.1, p. 5122, defines congruence arithmetic Fuchsian groups; proof of Corollary 1.2, equation (5.2), p. 5126, records $\lambda_1 \ge 975/4096$ for every such group via Jacquet–Langlands and Kim–Sarnak.

25. Hervé Jacquet and Robert P. Langlands, *Automorphic Forms on* GL(2), Lecture Notes in Mathematics 114, Springer, 1970. DOI, IAS-hosted text. Chapter III, especially §16, supplies the global quaternionic transfer; at split places, including the real place here, the local component is preserved.

26. John Voight, *Quaternion Algebras*, Graduate Texts in Mathematics 288, Springer, 2021. DOI, open author version. Proposition 15.5.2 (maximal superorders); §23.1.3 (Eichler orders at ramified primes); equations (27.6.6)–(27.6.7) (compact-open finite adelic order/unit group); Theorem 28.2.10 (strong approximation for indefinite B^1); Definition 38.7.2 (principal congruence groups).

27. Jiamin Li and Jianya Liu, “Anisotropic quadratic equations in three variables.”

    arXiv:2501.15033v2; Theorems 1.1–1.2, pp. 2–3; Lemma 2.3, pp. 5–6. The global form-level hypothesis is $t\det(f)$ square-free; the quantitative input is a smooth radial, congruence-orbit estimate with

$$O_\varepsilon\left(d^{1+\varepsilon}T^{1/2+\theta+\varepsilon}\right).$$

28. David Lowry-Duda, *On Some Variants of the Gauss Circle Problem*, arXiv:1704.02376. Theorem 5.5.4, thesis pp. 97–98.

14.12 Audit limitations

The claim-by-claim source versions and locators used in this bounded audit are enumerated in section 14.11. The canonical tree does not preserve byte-identical snapshots or retrieval hashes for those external publications, so this file does not assert a reproducible local source

corpus. Good's 1983 book was the sole exception to direct full-text access: its theorem locator and statement were checked through the explicit restatement and independent proof in Chatzakos–Petridis (2016) and the version-dated Lekkas–Petridis treatment.

The statement that no theorem was located is a result about the source versions enumerated here, not a proof that no unpublished or differently indexed result exists. It does establish that none of the enumerated works may be cited as a proof of the exact unconditional square-root discrepancy.

15 Interpretation and limits

Five logically distinct facts should not be merged. First, the Koide equation is a null-cone equation and has an exact determinant and phase model. Second, the defect-one integral shell admits Lorentzian reflection dynamics, unique reduced roots, exact ternary coordinates, and explicit Pell families on its chamber faces. Third, its HNF ideal data and positive-unit powers give an exact parametrization, inverse, multiplicity-one statement, sharp floor formula, and shifted Hecke-phase identity. Fourth, homogeneous dynamics and arithmetic invariant measures give the explicit main coefficient, while direct effective well-roundedness and Benoist–Oh give an unspecified power saving. Fifth, the exact quaternion shell bijection produces finitely many primitive hyperbolic conjugacy classes, but the radial Huber height is not the unchanged Koide height. The smooth radial 39/64 and sharp radial 2/3 theorems therefore do not prove the open pointwise five-wall square-root estimate. None of these statements explains why measured charged-lepton masses should obey the Koide relation.

The S^6 occurrence is still weaker: Proposition 1.3 identifies coefficient spaces, not global moduli. Corollary 1.5 shows that a point of the Koide cone gives a singular real period matrix for every choice of real parts, whereas (1.24) places the recorded positive radial cusp matrices on the nonzero-determinant side for all sufficiently large T. The full Koide cone is the discriminant hypersurface of the ambient coefficient model; its nonnegative-mass image lies in the rank-one boundary of the positive-semidefinite cone. Neither is a family of smooth fibres supplied by the manuscript.

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